# What Do Audit Trails Buy?

# Learning Trading Algorithms from Prices

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**Abstract**

Regulators increasingly require firms to identify the algorithm behind every order, as under the U.S. Consolidated Audit Trail and MiFID II. We ask what a supervisor could learn about trading algorithms from public prices and order flow alone, and what disclosure buys instead. In a closed-loop market where algorithms trade on exponentially weighted price signals, prices reveal only a merged summary of the algorithms. Separating two algorithms whose memories differ by $\delta$ costs trading capital of order $\delta^{-4}$; separating $K$ unknown algorithms requires $\Theta((\Delta\eta)^{-(4K-2)})$ experiments, and with general memory accuracy improves only as $1/\log N$. Passive data face the same barrier. Class-tagged disclosure removes it, saving a factor of order $(\Delta\eta)^{-(4K-4)}$ in samples. At calibrated parameters, learning two unknown algorithms from prices costs between $15 million and $356 million a year per market in trading losses. In data, the trader-class reports that the CFTC mandates restrict memory that prices leave unrestricted in all eight regulated U.S. futures markets in a registered test, as do Binance's class ratios in 29 of 29 fresh markets.

**Keywords:** system identification, market microstructure, algorithmic trading, minimax sample complexity, moment problems, value of disclosure, transient risk.

**JEL:** C14, C32, C58, D83, G14.

**Reproducibility.** Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix I maps each table, figure and number to its source. The specifications of the October 2026 follow-up tests (Appendices F and 6.4) were committed to the repository before their test data were downloaded; because each specification reached the public repository in the same push as its results, that order is documented by the commit history rather than by an independent timestamp.

# 1 Introduction

Regulators increasingly require firms to say which algorithm sent each order. Under MiFID II, EU investment firms must flag algorithmic orders and identify the algorithm responsible for each investment and execution decision in their transaction reports (ESMA, 2016); the US Consolidated Audit Trail (CAT) records who stands behind every order, at an operating budget of about $228 million a year (SEC, 2012, 2024). As trading passes to learning algorithms and AI agents (Colliard et al., 2022; Dou et al., 2025), the question behind these regimes becomes pressing: could a supervisor learn what the algorithms do from public prices and order flow, or does it need the audit trail? This paper answers that question and puts a price on the answer.

The answer is that prices cannot reveal the algorithms at any affordable cost, and class-level disclosure can.

The reason is the feedback loop. Algorithms update on recent prices, their orders move prices, and prices feed back into the algorithms. We model $K$ cohorts of algorithms that track the price with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Lemma 2.1). Each algorithm reaches the price only through the summed response of all of them, so prices identify a merged summary of the algorithms and nothing more (Theorem 3.1). Two algorithms with nearby memories leave nearly the same price path, and telling them apart requires trading against them, with probes that move the price against the prober. Separating memories that differ by $\delta$ needs executed quantity of order $\delta^{-2}$ and trading capital of order $\delta^{-4}$ (Theorems 4.1 and 4.3). Separating $K$ unknown algorithms needs a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ (Theorems 5.4 and 5.5); with general memory, accuracy improves only as $1/\log N$, so halving the error squares the number of experiments (Theorem 5.7). Watching instead of trading does not help: passive data face the same polynomial barrier (Theorem 5.9).

In money the barrier is large. At calibrated parameters, re-learning two unknown algorithms each week from prices alone costs between $15 million and $356 million a year per

market in round-trip trading losses, against about $228 million a year for the entire Consolidated Audit Trail; with three unknown algorithms it exceeds $40 million per market even when their memories are far apart (Section 6.2). These are lower bounds on what learning from prices costs, not estimates of social cost.

Disclosure removes the barrier. When each class's flow is tagged, every class becomes an ordinary estimation problem whose precision does not depend on how similar the algorithms are (Proposition 6.1), and tagging saves a factor of order $(\Delta\eta)^{-(4K-4)}$ in samples (Theorem 6.2).

Two disclosure regimes let us check this prediction in data. The U.S. Commodity Futures Trading Commission requires large traders to report their positions and publishes them weekly by class. In a test registered before any data were downloaded, in eight regulated futures markets (CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, crude oil and natural gas), the class reports pin the memory of the faster class (leveraged funds or managed money) to an interval 9 to 71 times narrower than weekly prices allow, while prices leave both memories anywhere in the admissible range (Section 6.3). Binance's five-minute trader-class ratios show the same in every one of 30 perpetual futures markets and, in a second registered test on 30 markets never used before, in 29 of 29 (Section 6.4). The tests also produced results we did not predict, and we report them: leveraged funds sell rather than buy after price rises in three financial futures markets, and the Binance class persistence, once corrected for a within-day bias, is close to a random walk. The data therefore show what prices cannot bound, not that class persistence is learning memory.

Two further results complete the picture. Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient losses, so a supervisor cannot replace identification by monitoring eigenvalues or volatility (Theorem G.1). And standard reduced-form feedback indices computed from public data measure mostly filtering mechanics and clock alignment: two fee changes that moved trading volume

by factors of 2.6 and 4.9 left measured feedback unchanged (Appendix H).

The contribution is a price for transparency. A large literature asks how transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask what disclosure lets a supervisor learn about the algorithms that trade, and answer in samples, in trading capital and in dollars, with matching upper and lower bounds.

**What we do not claim.** The identification results hold for the stated linear model and observation contracts. They are not universal limits on surveillance with participant-level records, which can be far more informative (Propositions 4.4 and 6.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-registered real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Section 6.4). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 6.1, not an instance of it.

## 1.1 Related literature

**Algorithms, AI and supervision.** Recent work in finance asks how learning algorithms and AI agents behave in markets: reinforcement-learning market makers and informed traders can sustain wide spreads or collusive outcomes (Colliard et al., 2022; Dou et al., 2025), prices formed against Q-learners have distinctive dynamics (Banerjee and Szydlowski, 2026), and algorithmic markets can be manipulated (Tremacoldi-Rossi, 2025). These papers model or simulate the algorithms. We take the supervisor's side and ask what can be learned about them from market data, and what disclosure is worth. A related literature measures

the cost of regulation and of regulatory reporting (Trebbi et al., 2024); our dollar comparison with the Consolidated Audit Trail is a statement about the benefit side: what the audit trail buys that prices cannot.

**Price formation and impact.** Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Alfonsi et al. (2016) show that absence of price manipulation requires positive-definite decay kernels, which is the property Lemma 4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Appendix H. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

**Algorithmic trading and feedback.** Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

**Learning algorithms and what can be inferred about them.** Empirical game-theoretic analysis treats the choice among simulated trading strategies as a game (Wellman, 2006). Empirically, algorithmic trading has been found to improve liquidity and price discovery on average (Hendershott et al., 2011; Brogaard et al., 2014). These studies simulate the algorithms or observe them through regulatory data; we ask what an outside observer

can infer about them from market data alone, and answer in samples and in capital. Our stance is that of partial identification (Manski, 2003): when the structure is not identified, characterize what the data do determine, here the merged measure and, through it, the intensity-weighted distribution of the desks' information quality (Proposition 4.6).

**Control without identification.** We do not claim that acting on a system without identifying it is new. Moallemi et al. (2008) study strategic learning about a hidden position from prices, and van Waarde et al. (2020) characterize when control is possible without unique system identification. Our results concern a specific market observation contract and its costs in executed quantity, capital and money.

**Closest work.** Three literatures are closest. Super-resolution and moment-based mixture estimation give rates for recovering clustered point sources and mixing measures from noisy moments (Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our $K$-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for $K$ clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table 1): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 3.1); second, passive price observations incur the exact same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 5.8 and 5.9); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^\top T_h v$ (Theorems 4.1–D.4); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$ (Proposition 6.1 and Theorem 6.2).

Table 1: Comparison with finite-mixture estimation and open-loop identification

| Feature | Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020) | Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021) | Closed-loop market identification (this paper) |
|---|---|---|---|
| Where rates live | Component locations / moments of density | Poles of impulse response | Transmission zeros of closed-loop $H(z)$ (Thm 3.1) |
| Active vs. passive data | i.i.d. draws only | Designed input vs. output spectrum | Same $\Theta((\Delta\eta)^{-(4K-2)})$ order via minimum-phase Whittle gap (Thm 5.9) |
| Economic unit of cost | Sample size $N$ | Signal-to-noise ratio | Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 4.1–D.4) |
| Effect of class tagging | Observed component labels ($\Theta(1)$) | Separate sensor per mode | Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-4)})$ (Thm 6.2) |

Beyond the $K$-cohort case, we prove matching lower and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem 5.7) and price the resulting disclosure savings against audit-trail budgets.

**Statistical identification.** The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 5 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and Wu and Yang (2020), and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, and the result that the relevant rates are zeros, not poles, of the observable transfer function.

**Non-normal dynamics and supervision.** Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem G.1), which motivates the identification-free control of the companion paper. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 6.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

## 1.2 Outline

Section 2 sets out the model and Section 3 the identification boundary. Section 4 turns memory resolution into executed quantity, capital and money for a known pair, and Section 5 treats unknown mixtures, general memory and passive data. Section 6 states what disclosure buys, prices it against the Consolidated Audit Trail, and tests it on the CFTC's mandated trader-class reports and on Binance's class ratios. Section 7 draws the implications for supervision and Section 8 concludes. Proofs are in Appendix A; the internet appendix contains account-level capital, near-colliding rates, robustness to learning rules, the spectral blind spot and the measurement of reduced-form feedback, and Appendix I maps every

Table 2: Map of results

| Question | Answer | Where |
|---|---|---|
| What do prices identify? | The merged rate–weight measure only | Thm 3.1 |
| Quantity to separate a known pair | $\Theta(\delta^{-2})$, exact error formula | Thm 4.1 |
| Capital to separate a known pair | $\Theta(\delta^{-4})$, counterparty or own funds | Thm 4.3, Lem. 4.5 |
| Constant at the calibration | $[61.41, 64.48]$ par units; $\approx 10^5$ USDT | Thms D.2–D.4, Table 20 |
| Does adaptivity help? | At most by a factor $\approx 0.49$ in expected loss | Thm D.3 |
| Unknown mixtures | $(\Delta\eta)^{-(4K-2)}$, tight for every $K$ | Thms 5.4, 5.5 |
| General memory | Accuracy $\asymp 1/\log N$, lower and upper bounds | Lem. 5.6, Thm 5.7 |
| Passive data | Identifies $M$; same polynomial cost for every $K$ | Thms 5.8, 5.9, Prop. C.2 |
| Nonlinear and adaptive rules? | Settled online learners obey $\delta^{-4}$; fast adaptation or thresholds break it | Table 22 |
| What removes the barrier? | Tagged flow, audited equity | Props 4.4, 6.1 |
| What is tagging worth? | Factor $(\Delta\eta)^{-(4K-4)}$; $\approx 333\times$ for BTCUSDT (illustrative) | Thm 6.2 |
| Disclosure in dollars? | Annual probing loss exceeds CAT budget once $K \geq 3$ or $\Delta\eta \leq 0.04$ | Table 11 |
| Do class data restrict class persistence? | Yes: 30 of 30, and 29 of 29 new markets (registered); whether it is memory is open | Table 13 |
| Can eigenvalues certify safety? | No | Thm G.1 |
| What do feedback indices measure? | Mostly filtering and timing; small real feedback | Tables 24, 26 |
| Does feedback track activity? | No: fees move volume 2.6–4.9×, not $\hat{Q}$ | Table 28 |

number to its source.

## 2 Model and microfoundations

Time is discrete. Let $p_t$ be the price, $p^*$ the fundamental value and $x_t = p_t - p^*$ the pricing error. Competitive dealers absorb net order flow with price impact $\kappa > 0$, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where $v_t$ is an executed probe quantity and $w_t$ a price innovation. Cohort $i$ holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j \geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z - a) D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1 - S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), $J$ is entrywise nonnegative, its price row sums to $S$, and the economy is Schur stable iff $S < 1$.

**Lemma 2.1** (Filtering and learning microfoundations)**.** *(i)* Kalman memory. *A desk that*

*tracks a latent random-walk trend* $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, *from signals* $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, *using the steady-state Kalman filter, updates as* $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ *with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{7}$$

*(ii)* Gradient memory. *Online gradient descent with* $L_2$ *regularization* $\gamma \in (0,1)$, *step size* $\alpha > 0$ *and loss* $\frac{1}{2}H(x_t - \theta_t)^2$ *gives* $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ *with* $\eta = \gamma + \alpha H \in (0,1)$ *and* $q = \alpha H/\eta$. *Tabular Q-learning with learning rate* $\eta$ *has the same form.*

*Remark* 2.2 (Why exponential memory). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk $i$ chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1 - \delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \le \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i} u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i} I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1 - \delta_{\text{inv}})/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one (`microfoundation_and_baselines.py`). This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in the companion paper). The memory rate encodes the desk's signal-to-noise ratio or regularization, which is the private information a supervisor would like to recover; Section 3 shows why prices reveal it only in merged form.

**Lemma 2.3** (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \ge 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \le \lambda_\ell < 1.$$

*It is completely monotone and nonincreasing; its peak is the direct impact $\kappa$, even though $J$ is non-normal. In price/memory coordinates the price-only gain is* $\sup_t \|e_x^\top J^t\|_2 = 1$.

Numerically, $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.3 matters twice: it makes probing costs positive (Section 4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

## 2.1 Discussion of the assumptions

**Linearity.** The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

**Price-first timing.** We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

**Exponential memory.** Lemma 2.1 justifies exponential memory for filtering and regularized learning. Neural and reinforcement-learning policies are not exponential filters. Rules whose orders are smooth functions of an exponential-memory state are exponential filters to first order, so the results hold for small probes; Appendix F checks this in stylized simulations and finds where it fails, and how closely production algorithms follow such rules remains an empirical question we do not settle. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of cohorts, for which Theorem 5.7 shows that accuracy improves only logarithmically in the

data.

**Algorithms that react to being probed.** The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not separate them, and an algorithm that hides by mimicking another rate makes the hypotheses closer, so neither lowers the costs we derive; a reaction that differs across cohorts is itself a signature, but one the observer must model, which is a new identification problem rather than a free one. Second, the passive route needs no probe: Theorem 5.9 shows that natural price variation identifies the same separation at the same polynomial order in sample length, so an observer who fears detection pays in time, not in rate. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Appendix D trades at most 1.25 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

**Known primitives.** The resource theorems take $\kappa$, $a$ and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid, while the upper bounds require estimating them, which is cheap relative to the memory rates because they enter at first order.

**Observation contracts.** Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

# 3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

**Theorem 3.1** (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state $x_t$.*

*(i)* Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1). \tag{8}$$

*For $a > 0$ the minimal realization has dimension $1+|\mathrm{supp}(M)|$, counting distinct rates with nonzero merged weight after signed cancellation; for $a = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M)|$. A cohort with $\eta = 1$ merges into the anchor $a$ without adding a pole.*

*(ii)* Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with rates bounded away from 0 and 1. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{K^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

*where $K$ depends only on the family.*

The memory rates are poles of $G$ and *zeros* of $H$, not closed-loop poles. This has a practical consequence we verify in Section 5: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$,

$a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

*Remark* 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The $\delta^{-4}$ statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

# 4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, $\Phi$ is the standard normal distribution function.

## 4.1 Quantity

**Theorem 4.1** (Sharp probing-resource orders for a known symmetric pair)**.** *Fix* $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ *and* $\eta \in (0, 1)$. *Compare a single cohort of weight* $q$ *and rate* $\eta$ *with two cohorts of weight* $q/2$ *and rates* $\eta \pm \delta$, *both in* $[\epsilon, 1 - \epsilon]$. *All parameters are known. The price receives executed quantity through* $\kappa v_t$, *starts from zero, has no process noise, and is observed with independent* $N(0, \sigma^2)$ *noise. Probes are deterministic and common to both alternatives. Let* $d_T = (h_j^\delta - h_j^0)_{j<T}$ *and* $K_* = 2\kappa^2 q/[\epsilon^3(1 - S)^2]$.

*(i) For every probe with* $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{10}$$

*(ii) The optimal worst-case error over these probes and all classifiers is exactly*

$$e_T^*(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{11}$$

*attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed $V$. Error at most $\alpha < 1/2$ requires $V \geq 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of $T$.*

*(iii) The first two taps coincide and $d_2 = -\kappa^2 q\delta^2$. A single pulse of size*

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q\delta^2} \tag{12}$$

*and the third observation achieve error exactly $\alpha$. The necessary and sufficient quantity order is $\delta^{-2}$.*

*(iv) Under average power $\sum_{t<T} v_t^2 \leq PT$, the necessary horizon order is $\delta^{-4}$, attained by the pulse whenever $T \geq \max\{3, \lceil A^2/P \rceil\}$.*

*(v) For a fixed monetary price exposure $\lambda \neq 0$, additive reserve $L$ and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.*

The identity in (iii) is verified to machine precision for $\delta \in \{0.02, 0.05, 0.10\}$: the difference is $-1.2 \times 10^{-4}$, $-7.5 \times 10^{-4}$ and $-3.0 \times 10^{-3}$ at $\kappa = 1$, $q = 0.3$, and $-3.2 \times 10^{-4}$ at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 5).

**A Monte Carlo check of the exact error.** Formula (11) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table 3); simulated errors match the formula to Monte Carlo precision under both hypotheses.

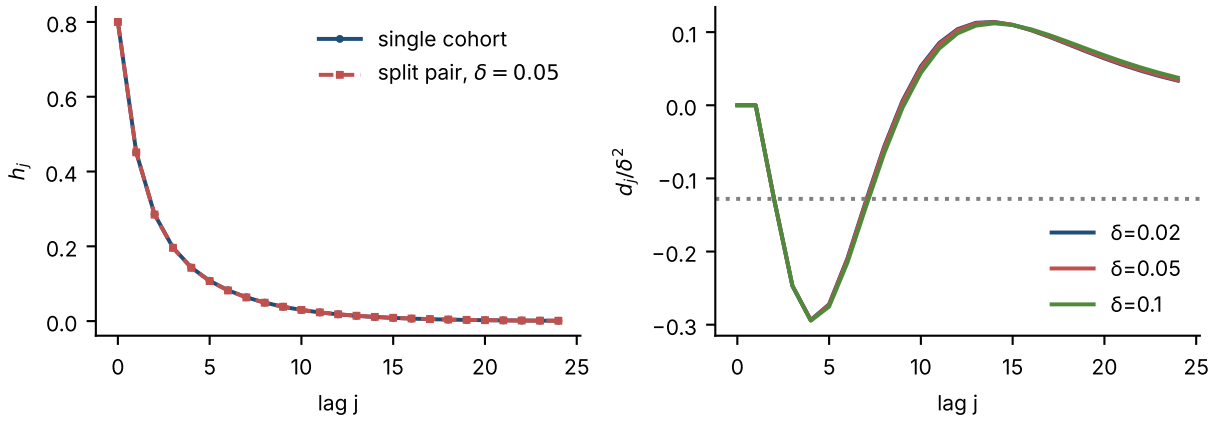



Figure 1: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by $\delta^2$ collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Table 3: Exact known-pair error: formula versus simulation

| Pulse $V$ | Horizon $T$ | $\Phi(-V\|d_T\|_2/2\sigma)$ | Simulated error, $H_0$ | Simulated error, $H_\delta$ |
|---|---|---|---|---|
| 20 | 3 | 0.3745 | 0.3734 | 0.3762 |
| 40 | 3 | 0.2611 | 0.2599 | 0.2603 |
| 40 | 10 | 0.0029 | 0.0030 | 0.0027 |
| 80 | 20 | $1.8 \times 10^{-10}$ | 0 | 0 |

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Figure 1 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to $\delta^2$, and the whole difference sequence scales with $\delta^2$. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like $\delta^{-2}$, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like $\delta^{-4}$.

## 4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and $\kappa$ currency per squared quantity.

**Lemma 4.2** (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant funda-*

*mental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{13}$$

*In the positive stable economy $T_h$ is positive definite, so no nonzero round trip produces a deterministic gain.*

We verify the ledger identity (13) against an explicit cash account to $5.7 \times 10^{-13}$ on 1,000 random probes, and the smallest eigenvalue of $T_h$ over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

## 4.3 Capital supplied by a counterparty

**Theorem 4.3** (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{14}$$

*equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least $\delta^{-4}$, and a single positive pulse attains that order.*

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure 2 illustrates the experiment.

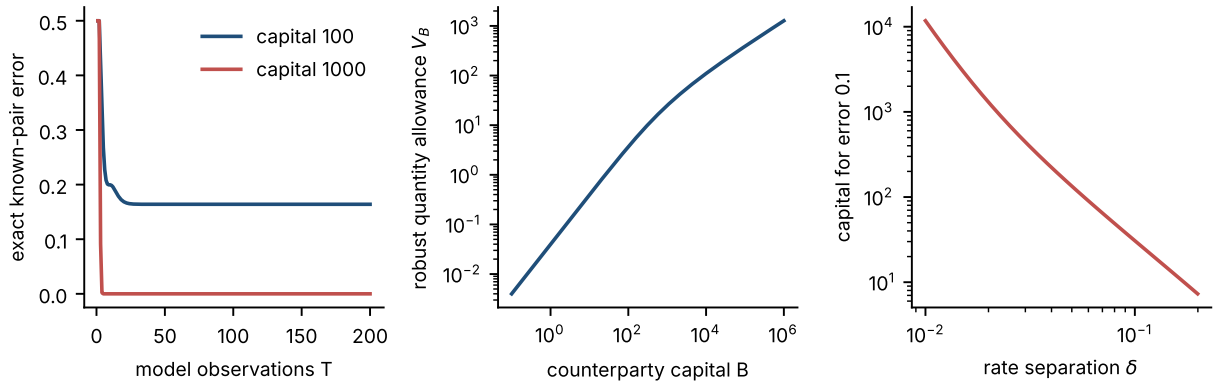



Figure 2: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (14). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

**Proposition 4.4** (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse* $0 < A \le V_B$ *distinguishes the known pair with zero error for every* $\delta > 0$.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 6).

## 4.4 Capital paid by the probe trader

**Lemma 4.5** (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash* $B > 0$ *and zero inventory; a finite signed sequence* $v$ *changes inventory by* $+v_t$ *and cash by* $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, *after which trading stops and the trader must remain solvent as the price returns to* $P_*$. *Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

*Every solvent strategy satisfies* $D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le K_*^2 B\delta^4/(\sigma^2 m)$*, so error at most* $\alpha < 1/2$ *requires*

$$B \;\ge\; \frac{2\sigma^2 m(1-2\alpha)^2}{K_*^2\delta^4}, \tag{15}$$

*without a quantity cap and at any horizon. Conversely, for any margin fraction* $0 < \mu \le 1$*, the pulse* $A$ *in* (12) *achieves error* $\alpha$ *and remains margin-solvent at every later mark with capital* $B_A = \frac{\kappa}{2}A^2 + (c + \mu P_*)A$*. Necessary and sufficient capital orders are* $\delta^{-4}$*.*

Lemma 4.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the $\delta^{-4}$ rate before the exact frequency-domain optimization of Appendix D. Its explicit lower constant is correspondingly loose: at the calibrated parameters it evaluates to $4.9 \times 10^{-4}$ par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any design we can execute, and because $m/K_*^2 \propto \epsilon^7(1-S)^5$ it collapses near the boundary of the family. We therefore state it as a lemma; Appendix D computes the constant that matters, within 5% of a feasible design at every separation.

## 4.5 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar\eta = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem 4.1 around $\bar\eta$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q\delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar\eta$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: by the $\delta^{-4}$ scaling of Theorem D.2, the optimal flat probe needs about $61.41 \times (0.05/0.0348)^4 \approx 262$ par units. Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly $4 \times 10^5$ USDT

and the naive pulse about $3 \times 10^7$ USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 5.4).

## 4.6 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (7) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\big(\sqrt{s^2+4s} - s\big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and $\eta$ is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

**Proposition 4.6** (Identification in economic primitives)**.** *Consider $K$ Kalman desks (Lemma 2.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$, and write $c_- = \eta'(\bar{s})$ and $c_+ = \eta'(\underline{s})$.*

*(i) Prices and aggregate order flow depend on the desks only through the measure*

$$\sum_i q_i\, \eta(s_i)\, \delta_{1-\eta(s_i)}.$$

*Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.*

*(ii) Fix a centre $s_c$ and let the ratios lie in $s_c + \Delta s\,[-1, 1] \subset [\underline{s}, \bar{s}]$. Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order $\Delta s$, or its weights $q_i\eta(s_i)$, from $N$ noisy reset experiments with worst-case error probability*

*below* $1/4$ *requires* $N \geq c\,\sigma^2(\Delta s)^{-(4K-2)}$*; on separated configurations, Prony's method of Theorem 5.5 composed with* $\eta^{-1}$ *achieves this order. The constants are those of Theorems 5.4 and 5.5 multiplied by powers of* $c_-$ *and* $c_+$*.*

*Proof.* (i) By Lemma 2.1(i) desk $i$ is a cohort with memory rate $1-\eta(s_i)$ and weight $q_i$, so Theorem 3.1 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through $q_i$. (ii) On $[\underline{s}, \bar{s}]$, $\eta$ is a $C^1$ bijection onto its image with $c_- \leq \eta' \leq c_+$. Lower bound: take the two economies of Theorem 5.4 with centre $r_c = 1-\eta(s_c)$ and rate spacing $\Delta\eta = c_-\Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1,1]$, so they are two populations of Kalman desks in the class, and the bound $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)}\sigma^2(\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because $\eta$ is $c_+$-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates, which Theorem 5.4 bounds below by a constant times $\Delta\eta = c_-\Delta s$. Upper bound: ratios in $s_c + \Delta s\,[-1,1]$ with spacing at least $\gamma\Delta s$ have gains in $\eta(s_c) + c_+\Delta s\,[-1,1]$ with spacing at least $c_-\gamma\Delta s$, a class of the form required by Theorem 5.5 with rescaled separation $\gamma c_-/c_+$; Prony's method recovers the rates and weights at that theorem's rate, and $\eta^{-1}$, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

**Proposition 4.7** (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio,* $q = \varphi(s)$ *with* $\varphi$ *positive and* $C^2$ *on* $[\underline{s}, \bar{s}]$ *(for a mean–variance desk,* $\varphi(s)$ *increases with* $s$*), and let* $n_i \geq 0$ *denote the mass of desks with ratio* $s_i$ *(a divisible population, not an integer desk count). Write* $s(\eta)$ *for the inverse gain*

*map and $w(\eta) = \varphi(s(\eta))\,\eta$ for the weight per desk as a function of its gain.*

*(i)* Unknown numbers of desks. *The merged measure is $\sum_i n_i\, w(\eta(s_i))\,\delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1-\eta(\bar{s}), 1-\eta(\underline{s})]$ is of this form, with $n_i = m_i/w(\eta_i)$ for an atom of mass $m_i$ at $1-\eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition 4.6(ii) holds unchanged.*

*(ii)* Known numbers of desks. *If the $n_i$ are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at $\bar{\eta} \pm \delta$ with $n$ desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2[w(\eta)(1-\eta)^m]_{\bar{\eta}} + o(\delta^2)$, so separating them needs $N \geq c\,\sigma^2\delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

*(iii)* Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map $T$ from configurations $(s_1, \ldots, s_K)$ to perceived ratios that is continuous on a box $\prod_i[\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition 4.6 apply at it.*

*Proof.* (i) Desk type $i$ is a cohort with gain $\eta_i = \eta(s_i)$, memory rate $1-\eta_i$ and total weight $n_i\varphi(s_i)$, so it contributes $n_i\varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1-\eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i/w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1-\eta)^m$; the tap difference is $n[g_m(\bar{\eta}+\delta) + g_m(\bar{\eta}-\delta) - 2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta}) + o(\delta^2)$, with $\sum_m \sup|g_m''|^2 < \infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over $N$ experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how

many desks of each kind there are, and even then the simplest separation problem keeps its $\delta^{-4}$ order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map $T$ from a model of the desks' learning.

### 4.7 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order $\delta^{-4}$ blocks (Internet Appendix, Proposition C.2), the same order as the passive Whittle bound of Section 5. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Anonymous, 2026); cohort non-identification alone does not establish that the worst relevant cost exceeds the informed one.

## 5 Unknown mixtures and passive data

Theorem 4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

### 5.1 Why naive pole fitting fails

Because the memory rates are zeros of $H$ (Theorem 3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - ah_t$; (2) form a Hankel matrix of $g$ and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of $G$; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Appendix C.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Appendix C.3); with

Table 4: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

| Scenario | $K$ | $N^*$ probes | $\sigma_{\text{crit}}$ | ML success ($f = 1$) | RMSE/CRB |
|---|---|---|---|---|---|
| $\eta = 0.30$ | 1 | 0.1 | $6.05 \times 10^{-3}$ | 94% | 1.02 |
| $\Delta\eta = 0.30$ | 2 | 47.8 | $2.31 \times 10^{-4}$ | 94% | 1.01 |
| $\Delta\eta = 0.20$ | 2 | 597 | $6.55 \times 10^{-5}$ | 96% | 0.98 |
| $\Delta\eta = 0.10$ ($\delta = 0.05$ pair) | 2 | 40,374 | $7.96 \times 10^{-6}$ | 93% | 1.03 |
| $\Delta\eta = 0.05$ | 2 | 2,620,320 | $9.88 \times 10^{-7}$ | 95% | 1.03 |
| spacing 0.10 | 3 | 17,349,791 | $3.84 \times 10^{-7}$ | 98% | 1.22 |
| spacing 0.20 | 3 | 27,038 | $9.73 \times 10^{-6}$ | 98% | 0.88 |

Tolerance $\pm 0.01$ at 95%. $\sigma_{\text{crit}}$ is the single-probe noise at which one probe suffices; operational single-probe noise is $1.6 \times 10^{-3}$. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at $\sigma_{\text{crit}}$; success means every weight within $\pm 0.01$. Fitted log–log slope of $N^*$ in $\Delta\eta$: $-6.04$.

noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

## 5.2 Local information: the Cramér–Rao benchmark

**Proposition 5.1** (Cramér–Rao benchmark)**.** *Let $\theta$ collect the weights $w_1, \dots, w_K$ and rates $\eta_1, \dots, \eta_K$. Suppose the impulse response $h_t(\theta)$, $t \le T$, is observed with i.i.d. Gaussian error from $N$ reset probes of size $V_{\text{probe}}$ against innovation noise $\sigma_p$, so the error has standard deviation $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. With Jacobian $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, any regular unbiased estimator has* $\mathrm{Var}(\hat{w}_k) \ge \sigma_h^2 [F_{\text{unit}}^{-1}]_{kk}$*, and a 95% interval of half-width* tol *requires*

$$N^* \ge \Big(\frac{1.96}{\text{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\text{probe}}}\Big)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{16}$$

*For $K = 2$ with spacing $\Delta\eta$, $[F_{\text{unit}}^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$, so $N^* = \Theta((\Delta\eta)^{-6})$.*

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table 4), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \le 50$) unbiased estimators cannot resolve close weights.

## 5.3 A global two-point bound for any estimator

Proposition 5.1 applies to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

**Theorem 5.2** (Global two-point bound and matching upper bound, $K = 2$)**.** *Fix total weight $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target weight separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$. Let $p_{\mp} = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{17}$$

*(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k \eta_k^m$ for $m \in \{0, 1, 2\}$, while*

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{18}$$

*Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.*

*(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight $q$, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,*

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{19}$$

*so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2 / \|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. The same bound holds without any labelling convention for the permutation-invariant loss*

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|\big)$$

Table 5: Two-point sample complexity for two-cohort weight recovery

| $\Delta\eta$ | $\Delta M_3$ | $\|\Delta h\|_2$ | $N^*_{\text{LeCam}}$ (95%) | $N^*_{\text{CRB}}$ |
|---|---|---|---|---|
| 0.30 | $2.67 \times 10^{-4}$ | $1.98 \times 10^{-3}$ | 7.1 | 47.8 |
| 0.20 | $7.92 \times 10^{-5}$ | $5.60 \times 10^{-4}$ | 88.2 | 597.0 |
| 0.14 | $2.72 \times 10^{-5}$ | $1.89 \times 10^{-4}$ | 775.3 | 5,267.9 |
| 0.10 | $9.90 \times 10^{-6}$ | $6.84 \times 10^{-5}$ | 5,927.6 | 40,374.2 |
| 0.08 | $5.07 \times 10^{-6}$ | $3.49 \times 10^{-5}$ | 22,740.5 | 155,048.9 |
| 0.06 | $2.14 \times 10^{-6}$ | $1.47 \times 10^{-5}$ | 128,333.0 | 875,732.3 |
| 0.05 | $1.24 \times 10^{-6}$ | $8.50 \times 10^{-6}$ | 383,859.1 | 2,620,321.2 |
| 0.04 | $6.34 \times 10^{-7}$ | $4.35 \times 10^{-6}$ | 1,466,361.5 | 10,012,662.4 |

*Worst-case failure at* $\Delta\eta = 0.10$, $N = 50$, *1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Recomputed columns from `recompute_theory.py`. Estimators re-implemented in `recompute_p1_estimators.py`: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least $\tau$.

*in place of* $\|\hat{w} - w(\theta)\|_\infty$, *whenever* $\Delta\eta \geq 2\tau q/(q + 2\tau)$, *because then* $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$.

*(iii) Conversely, on the known-midpoint family a linear estimator that projects the averaged impulse response on the normalized third-moment direction and inverts* (18) *has error at most* $|\hat{M}_3 - M_3|/(2p_-p_+(\Delta\eta)^3)$, *giving* $N = O(\tau^{-2}(\Delta\eta)^{-6}\log(1/\delta_{\text{fail}}))$. *On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1g_3 - g_2^2 = w_1w_2\eta_1\eta_2r_1r_2(\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i\eta_i r_i^m$, *and perturbation analysis gives the same order.*

We recomputed the two-point quantities exactly (Table 5): the moment gap equals $4\tau p_-p_+(\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table 5).

*Remark* 5.3 (The hard directions carry no risk). The two economies of Theorem 5.2 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa\sum_i q_i)$ is identical, 2.353 at the calibration of Table 5: the directions in which prices are least

informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Anonymous, 2026). Identification and safety ask different questions of the same data.

## 5.4 Many unknown cohorts

**Theorem 5.4** ($K$-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$ and total merged weight $W > 0$. There exist two $K$-cohort economies with rates in $r_c + \Delta\eta[-1,1]$, positive weights, and merged measures whose moments of order $0, \dots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from $N$ independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

*The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.*

The construction uses the $K$-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have $K$ atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem 5.2. Table 6 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with $K$ on top of the exponent.

Table 7 and Figure 3 confirm the order numerically.

Table 6: Gauss and Gauss–Radau rules on $[-1, 1]$ used in Theorem 5.4

| $K$ | Gauss nodes / Gauss–Radau nodes | Max gap, orders $< 2K-1$ | Gap at order $2K-1$ |
|---|---|---|---|
| 2 | $\pm 0.5774$ / $-1$, 0.3333 | $6 \times 10^{-17}$ | 0.2222 |
| 3 | 0, $\pm 0.7746$ / $-1$, $-0.2899$, 0.6899 | $2 \times 10^{-16}$ | 0.0533 |
| 4 | $\pm 0.3400$, $\pm 0.8611$ / $-1$, $-0.5753$, 0.1811, 0.8228 | $4 \times 10^{-16}$ | 0.0131 |

All Gauss–Radau weights are positive.

Table 7: Numerical check of Theorem 5.4

| | $K=2$ | $K=3$ | $K=4$ |
|---|---|---|---|
| Theoretical exponent of $N^*$ | $-6$ | $-10$ | $-14$ |
| Fitted log–log slope (small $\Delta\eta$) | $-5.96$ | $-9.96$ | $-14.01$ |

Le Cam bound $N^* \propto \sigma^2/\|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

**Theorem 5.5** (Matching upper bound for every $K$)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i \neq j} |s_i - s_j| \geq \gamma > 0$, and weights in $[\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of $K$-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and $N$ independent reset experiments whose responses carry Gaussian noise with variance $\sigma^2$. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2} (\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(2/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(2K/\beta)}\;\epsilon_N$$

*with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence, and together with Theorem 5.4 the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.*

*Proof.* The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus $\kappa$ at $t = 0$) can be solved for $k_0, \ldots, k_{2K-1}$ from $h_0, \ldots, h_{2K}$ with $h_0 = \kappa \neq 0$; the map is smooth with Lipschitz constant

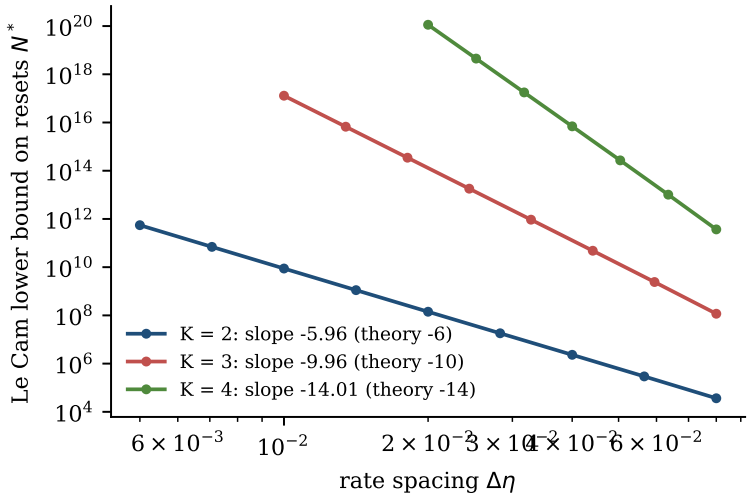



Figure 3: Two-point lower bound on the number of reset experiments needed to separate $K$-cohort economies, as a function of the rate spacing.

independent of $\Delta\eta$, so the estimated kernel values have error $O_P(\sigma N^{-1/2})$. Centring at $r_c$ is a fixed triangular linear map, $\mu_m = \sum_{j \le m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \ldots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. By the quantitative inverse function theorem there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F^{-1}$ exists and is $L$-Lipschitz on the ball of radius $\rho_0$ around every point of $F(\text{class})$. An error $e$ in $\mu_m$ is an error $e(\Delta\eta)^{-m}$ in $t_m$, largest at $m = 2K - 1$. The observed response errors are Gaussian; after nonlinear deconvolution the kernel errors need not be exactly Gaussian. On the compact family, a local Lipschitz bound for that smooth map and Gaussian concentration for the response coordinates give, for sufficiently large $N$, $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(2K/\beta)}\, \epsilon_N$ with probability at least $1 - \beta$. When this is below $\rho_0$, $\hat{t}$ lies in the ball where $F^{-1}$ is $L$-Lipschitz. The moment map is invariant to relabelling the atoms, so it is injective only on ordered nodes; on the class of ordered, $\gamma$-separated nodes it is injective, and the inverse function theorem applies there. Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound

Table 8: Sample requirement of Prony's method against the lower-bound order

| | $K=2$ | $K=3$ | $K=4$ | $K=5$ |
|---|---|---|---|---|
| Lower-bound order $-(4K-2)$ | $-6$ | $-10$ | $-14$ | $-18$ |
| Fitted slope of $N^*$ in $\Delta\eta$ (Prony) | $-5.94$ | $-10.00$ | $-13.93$ | $-18.34$ |

Rates $0.7+\Delta\eta\, s$ with $s$ the $K$-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; $N^*$ is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point.

below keeps unambiguous once $C\sqrt{\log(2K/\beta)}\,\epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \rho_0/C_1$. The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 5.4 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight $W$); for any such class the orders match, with constants that depend on the class. □

Table 8 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), here for real nodes, and it reproduces the ESPRIT slope of $-6.15$ reported for $K=2$ in the earlier version.

## 5.5 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m\, d\mu(r)$ with $\rho < 1$, where $\mu$ is any nonnegative measure of total mass $W$: a finite or infinite mixture of exponential memories with merged weight $W$. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a+\kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu,\mu') = \sup\Big\{ \int f\, d(\mu-\mu') :\ f \text{ 1-Lipschitz on } [0,\rho],\ f(0)=0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly $W$. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree $L$, and let $c_* = \inf_{L\geq 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table 9).

**Lemma 5.6** (Moment-matched cohort pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0,\rho]$, each with exactly $n$ atoms, whose moments of orders $0,\dots,2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \geq c_*\rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r-\rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Appendix A). Each measure is an $n$-cohort economy.

**Theorem 5.7** (Minimax rate for general memory)**.** *Let $N$ independent reset experiments observe the impulse response with Gaussian noise of variance $\sigma^2$ per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

*(i)* Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2W^2/((1-\rho)\rho)^2\big)$. Whenever $\Lambda_N \geq 4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu) \geq \frac{c_*\,\rho W\log(1/\rho)}{\Lambda_N}\Big) \;\geq\; \frac{1}{4e},$$

*and the bound already holds over $n$-cohort economies, for a finite $n$ that grows with $N$.*

*(ii)* Upper bound. *With $a$ and $\kappa$ known, invert the averaged response into kernel values $\hat{k}_0,\dots,\hat{k}_L$, with $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$, and let $\hat{\mu}$ be any measure on $[0,\rho]$ minimizing $\max_{j\leq L}\big|\int r^j d\hat{\mu} - \hat{k}_j\big|$ (a linear program). For every $\beta\in(0,1)$ there is $N_0$*

*depending only on* $(\rho, W, a, \kappa, \beta)$ *such that for* $N/\sigma^2 \geq N_0$,

$$\sup_{\mu \in \mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu}, \mu) > \frac{3\pi\rho W \log B(\rho)}{\log(N/\sigma^2)}\Big) \;\leq\; \beta.$$

*Hence the minimax accuracy is of exact order* $1/\log(N/\sigma^2)$*, and reaching accuracy* $w$ *requires and suffices* $N/\sigma^2 = \exp(\Theta(1/w))$.

*Proof sketch.* (i) Put the measures of Lemma 5.6, scaled to mass $W$, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int(1-rz)^{-1}d(\mu_A - \mu_B) = z^{2n-1}\int r^{2n-1}(1-rz)^{-1}d(\mu_A - \mu_B)$; on $|z| = 1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A - k^B\|_2$ by the same quantity. The response difference is $H_A H_B$ times the shifted kernel difference, so $\|h^A - h^B\|_2 \leq C_h\|k^A - k^B\|_2$. Choosing the smallest $n$ for which the Kullback–Leibler divergence of the $N$-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz $f$ with $f(0) = 0$, Jackson's theorem gives a polynomial $q$ of degree $L$ with $\sup|f - q| \leq \pi\rho/(4(L+1))$, and expanding $q$ in Chebyshev polynomials of $2r/\rho - 1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu}, \mu) \leq \pi\rho(W+\delta)/(2(L+1)) + 2\delta\rho B(\rho)^{L+1}$, where $\delta$ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1 - az - \kappa/H(z))/(\kappa z)$, whose error is at most $2((1+s)/\kappa)^2(L+2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2} L\sqrt{\log L})$. The choice of $L$ makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4} L\sqrt{\log L})$, negligible against the first term. Full proofs are in Appendix A. □

Theorem 5.7 replaces a polynomial barrier with a logarithmic one. With exponential memory and $K$ known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts and their spacing are also known. The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on

Table 9: General memory: moment-matched cohort pairs and the moment estimator

| | | | | | | |
|---|---|---|---|---|---|---|
| *A. Best approximation of* $\lvert x\rvert$ | | | | | | |
| $L$ | 2 | 4 | 6 | 10 | 20 | 30 |
| $L\,E_L(\lvert x\rvert)$ | 0.2500 | 0.2705 | 0.2756 | 0.2785 | 0.2797 | 0.2800 |
| *B. Pairs of Lemma 5.6* | | | | | | |
| Atoms per measure $n$ | 2 | 4 | 6 | 8 | 10 | 12 |
| $W_1(\mu_A,\mu_B)$, $\rho=0.9$ | 0.2250 | 0.1037 | 0.0669 | 0.0494 | 0.0391 | 0.0324 |
| $n\,W_1$, $\rho=0.9$ | 0.450 | 0.415 | 0.402 | 0.395 | 0.391 | 0.388 |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.9$ | $4.41\times10^{-2}$ | $1.03\times10^{-3}$ | $4.08\times10^{-5}$ | $1.96\times10^{-6}$ | $1.05\times10^{-7}$ | $5.95\times10^{-9}$ |
| $\lVert k^A-k^B\rVert_2$, $\rho=0.99$ | $7.41\times10^{-2}$ | $5.84\times10^{-3}$ | $1.07\times10^{-3}$ | $2.76\times10^{-4}$ | $8.33\times10^{-5}$ | $2.76\times10^{-5}$ |
| *C. Linear-programming estimator, continuous memory distribution,* $\rho=0.9$ | | | | | | |
| $\log_{10} N$ | 4 | 8 | 12 | 16 | 20 | 24 |
| Moments used $L$ | 2 | 3 | 5 | 6 | 7 | 8 |
| Error $d(\hat\mu,\mu)$ | 0.00878 | 0.00759 | 0.00432 | 0.00445 | 0.00382 | 0.00399 |
| Error $\times\log(N/\sigma^2)$ | 0.202 | 0.245 | 0.179 | 0.226 | 0.229 | 0.276 |

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to $10^{-61}$ (60-digit arithmetic); every $W_1$ exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; $L$ from the rule in Theorem 5.7(ii); measure fitted on a 401-point grid; 20 replications. Kind: recomputed (`general_minimax.py`).

how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho\to1$ the moment-matched pairs separate their kernels more slowly ($\rho^{2n-1}$ decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0,\rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table 9 checks the construction and the estimator. The pairs of Lemma 5.6 have exactly $n$ atoms, agree in all moments below order $2n-1$ to $10^{-61}$, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho=0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for $N$ from $10^4$ to $10^{24}$: a $10^{20}$-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

## 5.6 Passive data

**Theorem 5.8** (Passive identification: minimum phase)**.** *In a stable economy with $0 < a < 1$, distinct $0 < r_i < 1$ and nonzero merged weights, the zeros of $H$ are the memory rates $r_i$ together with a structural zero at $z = 0$, and $H$ is minimum phase. Degenerate anchor and zero-rate cases require the cancellations described in the merged-equivalence proof. With a single price innovation, the price spectrum identifies $H$ up to scale, hence the memory rates; with $\kappa$ and $a$ known, as in Theorem 3.1, it identifies the merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

**Theorem 5.9** (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of a single innovation with variance $\sigma^2 > 0$, for $\theta$ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

*(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.*

*(ii) For the moment-matched two-cohort economies of Theorem 5.2, $D = \Theta(\tau^2(\Delta\eta)^6)$.*

*(iii) For every $K \geq 2$, for the $K$-cohort Gauss and Gauss–Radau economies of Theorem 5.4, $D = \Theta((\Delta\eta)^{4K-2})$.*

*Hence any test based on $T$ passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, and the likelihood-*

*ratio test attains these orders as $T \to \infty$: passive data cost the same polynomial order as active probing for every number of cohorts.*

The proof (Appendix A) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order $\delta^2$ for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are $-4.21$ and $-5.93$, against the theoretical $-4$ and $-6$; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14 (`hardening/p1_checks.py`).

Passive surveillance is possible in principle but costs the same polynomial order as active probing. Proposition C.2 is the exact finite-sample version for the known pair.

# 6 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- $I_0$: public prices only.
- $I_1$: prices and aggregated executed flow.
- $I_2$: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- $I_3$: prices and exact audited equity (Proposition 4.4), also with reports rounded to 0.01.
- $I_4$: algorithm update timestamps without the update values.
- $I_5$: a supervisory audit registry with access to internal algorithm states.

Table 10 shows that disclosure is not one thing. Timestamps without update values carry no information about $\delta$ in this model. Noisy periodic statements barely help. Exact

Table 10: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

| Contract | Channel | Pulses | Benchmark | Saving vs. $T = 20$ |
|---|---|---|---|---|
| $I_0$: prices, $T = 3$ | public prices | 271 | 4,305.25 | −2,123.5% |
| $I_0^*$: prices, $T = 20$ | full price path | 12 | 193.62 | 0.0% |
| $I_1$: prices and signed flow | public flow | 12 | 193.62 | 0.0% |
| $I_2$: noisy equity ($\sigma_E = 0.5$) | statement | 12 | 193.62 | 0.0% |
| $I_3$: exact equity, 5-unit lot | audited equity | 1 | 12.70 | 93.4% |
| $I_3$: exact equity, 0.01 rounding | rounded report | 1 | 15.52 | 92.0% |
| $I_4$: timestamps only | no likelihood | 12 | 193.62 | 0.0% |
| $I_5$: audit registry | internal states | 0 | 0.00 | 100.0% |

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

**Proposition 6.1** (Tagged flow removes the barrier)**.** *Let the price be driven by i.i.d. innovations of variance $\sigma_\xi^2$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. measurement noise of variance $\sigma_f^2$ independent of prices. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)\big], \quad (20)$$

*where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s + m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (21)$$

*$V_i$ depends only on cohort $i$'s own $(w_i, r_i)$, $\sigma_f^2$ and the price autocovariances, which are bounded and nondegenerate uniformly on a uniformly stable family whatever the rate spacing;*

*so the merged weights are estimated at rate $T^{-1/2}$ with a constant that does not depend on $|r_i - r_j|$.*

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about $w_i$ does not involve the other cohorts' rates; the closed form (21) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with $4 \times 10^5$ periods; `hardening/p1_checks.py`). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between $5.8 \times 10^{-4}$ and $6.6 \times 10^{-4}$). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

## 6.1 How much disclosure is worth

Proposition 6.1 says that tagging removes the barrier; the next result says by how much.

**Theorem 6.2** (Value of class-tagged disclosure)**.** *In the class of Theorem 5.5, let $N_p$ be the number of experiments needed with prices alone and $N_t$ the number needed with each class's flow observed with noise $\sigma_f$. To recover the weights to fixed accuracy, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-2)}$; to resolve the rates to accuracy proportional to their spacing, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-4)}$.*

*Proof.* $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 5.4 and 5.5. With tagged flow each class is a single-exponential problem whose Fisher information for $(w_i, r_i)$ does not depend on the other classes, so $N_t \asymp \sigma_f^2$ for the weights and $N_t \asymp \sigma_f^2(\Delta\eta)^{-2}$ for rates resolved to a fraction of $\Delta\eta$. □

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of $-6.08$ for prices and $-4.34$ for the ratio at $K = 2$ (theory $-6$ and $-4$). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in

BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972). These rates come from day-demeaned regressions, which bias persistence down (Section 6.4), so the factors are illustrative.

## 6.2 Disclosure in dollars

Theorem 6.2 states the value of class-tagged disclosure in samples. Theorems D.2 and 5.4 let us translate that value into a flow of round-trip trading losses and set it beside the official operating budgets of the U.S. Consolidated Audit Trail (CAT). Two adjustments make the comparison exact and like-for-like:

(i) *Exact $K$-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem D.2,

$$L^*_{\mathrm{rt}}(K, \Delta\eta) \;=\; \frac{4\sigma^2 z^2_{1-\alpha}}{\sup_\omega\big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2/g_0(\omega)\big)},$$

directly for the $K$-point Gauss versus Gauss–Radau economies of Theorem 5.4 and Table 6, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss $L^*_{\mathrm{rt}}$ is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over and regulatory surveillance is periodic, we compare annual CAT operating costs against the annualised round-trip trading loss $52 \times L^*_{\mathrm{rt}}$ under weekly certification renewal (matching the CFTC weekly reporting cycle of Section 6.3). We benchmark against three primary official figures: the CAT LLC 2025 operating budget of $248.8 million in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to $228.3 million in May 2025; the SEC's 2016 CAT NMS Plan Approval Order estimate

Table 11: Price-only round-trip probing loss per market and annualised weekly-renewal cost against official CAT budgets

| | Loss / probe (USDT) | | Annual (52×, USDT) | | Mkts ($228.3M / $1.7B) | |
|---|---|---|---|---|---|---|
| Economy ($\Delta\eta$) | Roll | 1-sec | Roll | 1-sec | Roll | 1-sec |
| $K = 2$, known (0.20) | 6,200 | 149,901 | 322,400 | 7,794,852 | 708.1 / 5,273.0 | 29.29 / 218.09 |
| $K = 2$, known (0.10) | 99,192 | 2,398,422 | 5,157,984 | 124,717,944 | 44.3 / 329.6 | 1.83 / 13.63 |
| $K = 2$, known (0.04) | 3,874,696 | 93,688,369 | 201,484,192 | 4,871,795,188 | 1.1 / 8.4 | 0.05 / 0.35 |
| $K = 2$, Radau (0.20) | 4,292 | 103,788 | 223,184 | 5,396,976 | 1,022.9 / 7,617.0 | 42.30 / 314.99 |
| $K = 2$, Radau (0.10) | 283,488 | 6,854,612 | 14,741,376 | 356,439,824 | 15.5 / 115.3 | 0.64 / 4.77 |
| $K = 3$, Radau (0.20) | 804,403 | 19,450,107 | 41,828,956 | 1,011,405,564 | 5.5 / 40.6 | 0.23 / 1.68 |
| $K = 3$, Radau (0.10) | 1,137,325,880 | 27,500,017,407 | 59,140,945,760 | 1,430,000,905,164 | 0.0 / 0.0 | 0.00 / 0.00 |
| $K = 4$, Radau (0.20) | 156,632,086 | 3,787,291,895 | 8,144,868,472 | 196,939,178,540 | 0.0 / 0.2 | 0.00 / 0.01 |

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact $K$-cohort quadrature pair of Theorem 5.4 and Table 6 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem D.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. The final columns divide the revised CAT LLC 2025 operating budget ($228.3M; Nov. 2024 filing $248.8M, SEC Release No. 34-101901, p. 4) and the SEC 2016 annual industry estimate ($1.7B, Release No. 34-79318, p. 363) by the annual probing loss per market. Probing cost is a lower bound on the cost of learning from prices, not a social cost. Kind: recomputed (`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`).

of $2.4 billion in initial industry build costs and $1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating $4.0 billion initial and $2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

Probing cost is a lower bound on the cost of learning from prices, not a social cost. Within that interpretation, Table 11 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant $\Delta_K$ turn the comparison in favour of disclosure. For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly price-only renewal costs $0.32 million per market per year at Roll noise and $7.79 million at one-second noise, so matching the $228.3 million CAT LLC operating budget requires 708.1 or 29.29 markets. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs $14.74 million (Roll) or $356.44 million (one-second) per market per year, matching the

CAT LLC budget in 15.5 or 0.64 markets and the SEC's $1.7 billion industry figure in 115.3 or 4.77 markets. Once a supervisor must separate $K = 3$ unknown cohorts at $\Delta\eta = 0.20$, annual probing loss rises to $41.83 million (Roll) and $1.01 billion (one-second) per market (5.5 and 0.23 markets at the CAT LLC budget; 40.6 and 1.68 markets at $1.7 billion), and at $K = 3, \Delta\eta = 0.10$ or $K = 4, \Delta\eta = 0.20$ a single market's annual probing loss exceeds both the CAT LLC operating budget and the SEC's total annual industry estimate. The renewal frequency is an assumption: break-even market counts scale inversely with it, so monthly renewal multiplies every count in the last two columns by 4.3 and renewal on each of 252 trading days divides them by 4.8.

## 6.3 Mandated disclosure: the CFTC's trader-class reports

The cleanest test uses disclosure mandated by law. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes them weekly by class: leveraged funds, asset managers and dealers in financial futures, and managed money, producer/merchants and swap dealers in commodities. This is class-tagged disclosure mandated by law. We registered a test (SHA-256 `6f1d6f35ae88...`) before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \,\text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative fits a two-exponential kernel to twelve weekly return lags. Intervals come from a 13-week moving-block bootstrap. The registered hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table 12). The class reports pin each

Table 12: Trader-class memory from the CFTC Commitments of Traders, eight regulated futures markets

| Market | Weeks | Fast $\hat{r}$ [95% CI] | Slow $\hat{r}$ | $\hat{r}_{\text{fast}} - \hat{r}_{\text{slow}}$ [95% CI] | $\hat{w}_{\text{fast}}$ | Ratio |
|---|---|---|---|---|---|---|
| Bitcoin (CME) | 434 | 0.951 [0.928, 0.969] | 0.986 | −0.035 [−0.062, −0.010] | − | 23.7 |
| Ether (CME) | 282 | 0.890 [0.826, 0.932] | 0.960 | −0.070 [−0.143, −0.011] | − | 9.1 |
| E-mini S&P 500 | 1,039 | 0.914 [0.884, 0.937] | 0.954 | −0.040 [−0.071, −0.012] | − | 18.3 |
| 10-year T-note | 1,039 | 0.992 [0.985, 0.998] | 0.993 | −0.000 [−0.006, +0.007] | (0) | 71.4 |
| Euro | 1,035 | 0.961 [0.947, 0.972] | 0.996 | −0.034 [−0.051, −0.022] | + | 37.3 |
| Yen | 1,035 | 0.955 [0.939, 0.968] | 0.984 | −0.029 [−0.047, −0.013] | + | 34.0 |
| WTI crude oil | 1,039 | 0.961 [0.946, 0.974] | 0.999 | −0.037 [−0.053, −0.023] | + | 35.2 |
| Natural gas | 1,039 | 0.981 [0.969, 0.990] | 0.980 | +0.001 [−0.014, +0.017] | + | 44.0 |

*Pre-registered hypotheses*: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass.

Fast class: leveraged funds (financial futures) or managed money (commodities); slow class: asset managers or producer/merchants. $\hat{w}_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0, 0.99]$ in every market. Weekly returns from Binance spot (Bitcoin, Ether) and Yahoo Finance (others; the Treasury return is minus the change in the 10-year yield). Specification SHA-256 `6f1d6f35ae88...`; test opened once. Kind: real data (Part 11).

fast class's memory rate to an interval between 0.013 and 0.106 wide, while weekly prices leave both memory rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it: with prices alone the memories are not located at all. The classes differ as the cohort model allows. Leveraged funds and managed money have the shorter memory in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and producers; in Treasuries and natural gas the two memories are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class cannot represent. A post-hoc check is consistent with it but not decisive (Part 13, `R6`): weekly position changes of leveraged funds

and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and Treasuries ($-0.37$ to $-0.67$) but not in the euro or yen ($-0.04$ and $-0.05$), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal.

Together with Table 13, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions restrict persistence rates that prices leave unrestricted; whether that persistence is learning memory or slow turnover of positions, these data do not say.

## 6.4 Voluntary disclosure: Binance's trader-class ratios

Binance publishes voluntarily, for one venue but every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 6.1. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 6.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation $x_t$. For each class $c$ we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio $R$. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

Table 13 gives three results. First, tagged class data restrict persistence rates that prices

Table 13: Memory rates with and without class-tagged data

| Symbol | Days | Top position $\hat{r}$ [95% CI] | All accounts $\hat{r}$ [95% CI] | Width ratio | Classes differ |
|---|---|---|---|---|---|
| BTCUSDT | 48 | 0.9889 [0.9842, 0.9935] | 0.9986 [0.9955, 1.0005] | 121.7 | yes |
| ETHUSDT | 48 | 0.9793 [0.9616, 0.9907] | 0.9972 [0.9934, 0.9997] | 55.6 | yes |
| CELRUSDT | 48 | 0.9884 [0.9830, 0.9922] | 0.9882 [0.9804, 0.9914] | 81.6 | no |
| CVXUSDT | 33 | 0.9822 [0.9732, 0.9908] | 0.9829 [0.9694, 0.9900] | 45.9 | no |

| *Pre-registered test on fresh days, 30 USD-M perpetuals, 2025–2026* | |
|---|---|
| (i) width ratio $> 10$ | 30 of 30 (range 27.6–124.0) |
| (ii) price-only rate interval wider than 0.25 | 30 of 30 (range 0.65–0.99) |
| (iii) merged response $\sum_m k_m$ identified | 5 of 30 |
| Classes differ | 11 of 30 |

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification registered before the fresh days were downloaded (SHA-256 `65acda361861...`); a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

and flow leave essentially unrestricted: with the one-hour price-only kernel, the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing. The ratios also overstate the gap for a second reason: the price-only kernel spans one hour (twelve five-minute lags), while the tagged half-lives are several hours, so the price-only fit cannot see memories of that length. An exploratory check on the 30 symbols and 105 days of the follow-up test refits the price-only kernel with 48 and 96 lags (four and eight hours) and compares both routes in one unit, the half-life. With 96 lags the price-only slow half-life (median 5.6 hours) reaches the tagged scale (median 4.2 hours). The tagged intervals remain narrower in all 30 symbols, but by a median factor of 7.5 in log half-life rather than 28 to 124, and the price-only interval still reaches a bound of the fit in 25 of 30. The

horizon-matched comparison was then registered as a primary hypothesis of a test on 30 symbols never used in this project, before its data were downloaded, and passed: the tagged interval was narrower in 29 of 29 evaluable symbols (median factor 7.3, smallest 2.6), and the price-only interval reached a bound of the fit in 26. A registered secondary of the same test qualifies what the tagged estimate measures. Day-demeaned regressions bias persistence down; corrected by a split-panel jackknife, the median tagged rate is 1.0008 per five minutes, and the corrected interval lies below one in only 7 of 87 class fits. Within a day the class ratios are close to a random walk, so the half-lives of a few hours reported here are largely produced by removing each day's mean. The evidence therefore supports a narrower claim than identification of class memory: positioning data restrict a persistence parameter that prices cannot bound. Whether that persistence reflects learning memory or slow turnover of positions, which theory's tagged contract (class-tagged flow) would distinguish, is not established by these data. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin symbols the classes are often indistinguishable. Third, the pre-registered test fails its 80% criterion, on the part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 5.8 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely. This is a failure of the prediction as registered, and we report it as one. A post hoc reading, not tested here, is that identification is an asymptotic statement and 42 days of passive thin-market data are far from that regime. Two follow-ups were registered before their data were downloaded, and each ran once. On 105 untouched days in the same 30 markets, price-only intervals were again at least ten times wider than class-tagged ones in all 30 symbols (median ratio 74.9 with the one-hour kernel), but the merged response was significantly different from zero in only 9 symbols, and pinning it down was at least ten times cheaper than separating the memory rates in only 10 of 30 (23 were required): that prediction failed.

On 58 further days, the advantage of tagging grew with the sample in 22 of 29 symbols, the number required, but the tagged intervals shrank more slowly than the parametric rate (20 of 29 in the predicted range, 22 required), consistent with drift in class persistence over 21 months. Both results are reported in full in the companion thesis.

To test directly whether the memory rates identified by class-tagged positions improve out-of-sample return predictions after large order-flow shocks, we pre-registered a dedicated out-of-sample test (`P1_disclosure_oos.json`, SHA-256 `f0182ce50978...`) before downloading the 22nd-of-month data for January 2025–September 2026 across 19 thin perpetuals (6,137 top-decile five-minute flow shocks on 2024 training days; 4,579 shocks on untouched 2025–2026 test days). On the training sample we estimate each symbol's two memory rates $\hat{r}_1, \hat{r}_2$ either from the class-tagged positions or from the price-only two-exponential fit, construct the filtered memory states $y_{k,t} = \hat{r}_k y_{k,t-1} + (1 - \hat{r}_k)\text{ret}_t$, and forecast the 30-minute post-shock cumulative return $R_{t+1\to t+6}$ out of sample relative to an AR(1) return-and-flow baseline. The primary hypothesis fails: on the untouched test shocks, the price-only memory states yield $\Delta R^2 = -0.199$ percentage points relative to the baseline, the class-tagged memory states yield $\Delta R^2 = -1.288$ points, and the tagged-minus-price-only difference is $\Delta R^2 = -1.089$ points (95% date-block bootstrap CI $[-2.605, +0.376]$, Clark–West $t = 0.96$, $p = 0.1684$). Likewise, adding the level and one-hour change of the top-trader and all-account positions to the hourly tail-risk controls of the companion paper leaves out-of-sample $R^2$ unchanged ($-0.19$ points, CI $[-0.99, 0.83]$).

After this test failed, we registered a follow-up (`P1_disclosure_state_oos.json`, SHA-256 `5136cc6b846d...`) on untouched 2nd-and-9th-of-month data for January 2025–September 2026 across the same 19 thin perpetuals (12,046 top-decile five-minute flow shocks on 24 training dates in 2024; 9,875 shocks on 42 test dates; test opened once). It asks the narrower question that Theorem 3.1 and Proposition 6.1 answer directly: can prices and flow track the class states themselves? A two-exponential price-and-flow filter ($\alpha_{\text{fast}} = 0.50$, $\alpha_{\text{slow}} = 0.10$ on five-minute returns and signed flow, plus shock return, shock

flow and symbol fixed effects) is compared with a filter that adds the current class states and their one-bar changes. Both registered primaries pass by a wide margin (Table 14): for the 30-minute-ahead divergence between top-trader and all-account positions, out-of-sample $R^2$ is $-4.8\%$ with prices and flow and 46.0% with class data; for the top-trader state it is 4.4% and 54.8%.

Table 14: Pre-registered out-of-sample cohort-state trajectory test on untouched 2nd-and-9th-of-month days (`P1_disclosure_state_oos`), 2025–2026

| 30-min post-shock target (19 thin) | $R^2_{\text{pf}}$ | $R^2_{\text{tag}}$ | $\Delta R^2$ (pts) [95% CI] | CW $t$ ($p$) | Res. |
|---|---|---|---|---|---|
| Prim. 1: Divergence $d_{t+6}$ (9,875 shocks) | −4.766 | 45.962 | +50.728 [+45.291, +55.918] | 21.27 ($< 10^{-16}$) | pass |
| Prim. 2: Top state $n_{\text{top},t+6}$ (9,875 shocks) | 4.441 | 54.845 | +50.404 [+47.226, +53.724] | 21.85 ($< 10^{-16}$) | pass |
| Sec.: Divergence $d_{t+6}$, all (210,144 bars) | −0.586 | 37.053 | +37.639 [+36.232, +39.107] | 72.75 ($< 10^{-16}$) | — |
| Sec.: Top state $n_{\text{top},t+6}$, all (210,144 bars) | 0.186 | 46.779 | +46.593 [+45.555, +47.643] | 80.44 ($< 10^{-16}$) | — |

Out-of-sample $R^2$ in percent. Trained on 24 dates in 2024 (12,046 shocks, 120,384 bars); evaluated once on 42 untouched test dates (2nd and 9th of each month, Jan 2025–Sep 2026); 2,000-draw date-block bootstrap 95% CIs; Holm $p < 10^{-16}$ across both primaries. Spec SHA-256 `5136cc6b846d...` (`R_P1_disclosure_state_oos.json`).

The gain is almost entirely persistence. In an exploratory decomposition on the same test shocks (`hardened_empirical_audits.py` → `R_hardened_empirical_audits.json`), an AR(1) on the current level alone accounts for 100.5% and 99.1% of the improvement, and when the target is the 30-minute *change* in positions the class data add −0.26 and +1.41 points. Because the class filter observes the series it forecasts, this is a weak test: it confirms that prices and flow cannot reconstruct where the classes are positioned, which Theorem 3.1 implies, but it does not show that class data reveal how their positions will move, and neither registered test finds that class data improve return forecasts.

# 7 Implications for supervision and market design

**What a supervisor can learn from public data.** Prices and anonymous flow reveal the merged measure $M$, the closed-loop impulse response and the reduced-form operator, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This

is not a failure of econometric technique: Theorem 5.2 holds for every estimator, biased or not.

**What probing costs.** At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about $10^5$ USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as $\delta^{-4}$ and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by $2^{4K-2}$ for $K$ unknown cohorts.

**What disclosure buys.** Disclosure regimes differ sharply in value (Table 10). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 6.1). This gives a quantitative argument for audit-trail regimes that record the participant class of each order, and a quantitative argument against relying on periodic statements for the same purpose. Calibrated to Binance's published class ratios, separating the memories of the top-trader group and of all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 6.2). The class reports that the CFTC already requires have the same property in regulated markets (Table 12), so the argument concerns the granularity and frequency of existing disclosure, not a new kind of reporting.

**When audit trails already exist.** The US Consolidated Audit Trail (SEC Rule 613) gives the SEC and the self-regulatory organizations a record of every order in NMS stocks and listed options from origination to execution or cancellation, linked to an anonymized customer identifier and to the broker-dealer that handled it; since 2025 it no longer collects the names, addresses and years of birth of natural persons (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three ways. First, they price what the record is worth for the question of who drives a feedback loop: participant-level records remove a barrier that otherwise costs round-trip trading loss of order $\delta^{-4}$ and samples of order $(\Delta\eta)^{-(4K-4)}$ (Proposition 6.1,

Theorem 6.2); set against the CAT LLC 2025 operating budget ($248.8 million in November 2024, revised to $228.3 million in May 2025) and the SEC's 2016 annual industry estimate ($1.7 billion), the annualised weekly-renewal probing loss that disclosure makes unnecessary is small for two well-separated classes and exceeds the entire CAT budget once $K \geq 3$ at $\Delta\eta \leq 0.10$ or $K = 4$ at $\Delta\eta = 0.20$ (Section 6.2, Table 11). For this purpose a stable account identifier is what matters and names are not needed, which is a quantitative reason to keep the account linkage while removing personal data. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so an observer with account-level records who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the same exponent; MiFID II's algorithm identifier (ESMA, 2016) is the closer analogue of the tagged flow in Proposition 6.1. Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not available to exchanges, other participants or researchers. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

**What to monitor instead.** Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem G.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and operator confidence sets, both of which are identified from public data. These are cheap where identification is expensive (Remark 5.3); the companion paper shows that the summed response is certified at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Anonymous, 2026).

**How to measure feedback.** Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and

(iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

# 8 Conclusion and an open problem

Prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order $\delta^{-2}$ and trading capital of order $\delta^{-4}$ for a known pair, with a certified constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for $K$ unknown cohorts, passively as well as actively; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights match, populations whose learned weights differ are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier, and its value can be stated in samples and in dollars: a sample factor of order $(\Delta\eta)^{-(4K-4)}$, about 333 for the trader classes Binance already publishes in BTCUSDT, and an annualised weekly-renewal probing loss that matches the $228.3 million CAT LLC operating budget across 15.5 (Roll) or 0.64 (one-second) markets for an unknown two-cohort Gauss–Radau pair at $\Delta\eta = 0.10$ and exceeds it in a single market once $K \geq 3$ at $\Delta\eta = 0.10$; the CFTC's class reports restrict persistence rates that prices leave unrestricted in all eight regulated markets we examined, while a pre-registered out-of-sample test finds that class-tagged memory states do not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points). At realistic noise the capital barrier for a moderate separation is about $10^5$ USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

This leaves an open question for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response $g_m = c^\top J^m b$, the operator $J$ and finite-sample confidence sets for it are identified from public data even when cohorts are not. The companion paper uses exactly this interface to build an identification-free pre-trade control.

**Data and code availability.** All data are public: Binance archives (https://data.binance.vision), verified against published checksums. The pre-registered specification of the disclosure test, its SHA-256 hash and its test-set unlock counter are in the replication package. Every number in this paper is produced by a notebook cell or script that writes a result file (Appendix I).

**Use of AI tools.** The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

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# A Proofs

**Lemma 2.1.** (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance $P$ solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (7), and

the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2}H(x_t - \theta)^2 = -H(x_t - \theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

**Lemma 2.3.** Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to $\kappa u$ and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of $J'$ gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0, 1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since $J$ is nonnegative with row sums at most one, every price row of $J^t$ has $\ell_1$, hence $\ell_2$, norm at most one, attained at $t = 0$.

**Theorem 3.1.** With zero initial states, $y_{i,t} = \sum_{s \le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s \le t} k_{t-s} x_s$ and every response is determined by $(a, \kappa, k)$, hence by $M$. Conversely, equal responses for all probes imply equal $H$, hence equal $G$, and partial fractions of $G$ identify the distinct poles with nonzero residues $w_i$, i.e. $M$. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \ne i}(z - r_j)$, as in (4); at a retained $r_i$ the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j \ne i}(r_i - r_j) \ne 0$ (memory rates are positive because $\eta_i < 1$), so no root of $D$ cancels. The remaining numerator factor $z$ cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\operatorname{supp}(M)|$ for $a > 0$ and $|\operatorname{supp}(M)|$ for $a = 0$ (numerically, the rank of the Hankel matrix of $(h_j)_{j \ge 1}$ is $K + 1$ and $K$ respectively for $K = 1, 2, 3$; `hardening/p1_checks.py`). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z - 1 + \bar{\eta})\big((z - 1 + \bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least $\epsilon$. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = H_\delta H_0 (G_\delta - G_0)$ gives an $\ell_2$ gain at most $K\delta^2$. The mean difference of a power-constrained probe has squared norm at most $K^2\delta^4 PT$; the equal-noise KL is at most $K^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

**Theorem 4.1.** In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1 - S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1 - S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_* \delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_* \delta^2 \|v\|_1$, and Gaussian KL with Pinsker gives (10). For (ii), the mean difference is a lower-triangular convolution matrix times $v$; each column is a shifted copy of a prefix of $d_T$, so $\|\mu_\delta - \mu_0\|_2 \leq V \|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters $h_2$ with factor $\kappa^2$ while $h_0, h_1$ coincide; a pulse $A$ separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the $\ell_2$ gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \leq |\lambda| \kappa V$, attained by a single pulse.

**Lemma 4.2.** After the uncontrolled update, the pre-probe displacement at step $j$ is $\sum_{i<j} h_{j-i} v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2} v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \leq \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

**Theorem 4.3.** Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \leq x_t \leq \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x_{j+1}^0 - \frac{\kappa}{2} \sum_{j<t} v_j^2 + c U_t$. Its

quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of $h$ bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (14). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse $A$ funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

**Proposition 4.4.** The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

**Lemma 4.5.** Marking residual inventory at $P_*$ gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum |v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1 - 2\lambda\cos\omega + \lambda^2) \ge (1-r)/(1+r)$, so $v^\top T_h v \ge m\|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (15). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory $A$; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

**Proposition D.1.** Parts (i) and (ii) follow from (22) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for $\delta$; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

**Theorems D.2 and D.3.** For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2}v^\top T_{h_i} v = \frac{1}{2\pi}\int g_i(\omega)|V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi}\int |\Delta\hat{H}|^2|V|^2 \le \sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \ge 2\sigma z_{1-\alpha}$ gives (26). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \ge \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0\sum_t(\Delta\mu_t)^2/(2\sigma^2) \le \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$,

which gives (28).

*Part (iv).* Let $G_\theta$ be the $T \times T$ lower-triangular Toeplitz section of $H_\theta/\kappa$, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \le \sup_\omega |H_\theta|/\kappa \le \bar{M}$ because $|H_\theta(e^{i\omega})| \le \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; $\kappa/H_\theta$ is analytic on and outside the unit circle because $H_\theta$ is minimum phase (Theorem 5.8), and Lemma 2.3 gives $\operatorname{Re} H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1-\lambda_\ell\cos\omega)/(1-2\lambda_\ell\cos\omega+\lambda_\ell^2) \ge \sum_\ell \omega_\ell/(1+\lambda_\ell) \ge \kappa/2$, so $\|G_\theta^{-1}\| \le 2$ and $\Sigma_\theta \succeq \sigma_{\text{eff}}^2 I$ (the smallest singular value of $G_\theta$ is at least 0.522 in 762 random positive economies and sections; `hardening/p1_checks.py`). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top \Sigma_1^{-1}\Delta\mu \le \|(H_1 - H_0)v\|_2^2/(2\sigma_{\text{eff}}^2)$. For the covariance part, the eigenvalues $\lambda_k$ of $A = \Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)$, and $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \le \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\text{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1 \Delta G^\top)$ and $\|\Delta G\|_F^2 \le T\|h^\delta - h^0\|_2^2/\kappa^2 \le T K_*^2\delta^4/\kappa^2$ (proof of Theorem 4.1) gives a covariance part of at most $C_w T\delta^4$. The data-processing inequality turns errors $(\alpha, \beta)$ into $D_{\text{KL}} \ge \operatorname{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (27).

**Theorem D.4.** Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the $G_i$-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \ge c_{\text{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \ge 0$; replacing $\theta$ by $\theta + \pi$ flips $d$ and $s_\theta$ and leaves (29) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\text{req}}/\lambda(\theta)$ and $D = L_i(v) - L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \ge c_{\text{req}} - \sum_{j \le L-3} \lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j \le L-3} c_j^2 - L_0(\theta) \;\ge\; \sum_{j \le L-3} \Big(1 - \frac{\lambda_j}{\lambda(\theta)}\Big) c_j^2 \;\ge\; \Big(1 - \frac{\lambda_{L-3}}{\lambda(\theta)}\Big) \sum_{j \le L-3} c_j^2 \;\ge 0,$$

and $r^2 \ge L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)} \sum_{j \le L-3} c_j^2 \ge L_0(\theta)\big(1 - \gamma(\theta) D\big)$. By Hölder's inequality with the sign vector $s_\theta$, $c_{\text{fee}}\|Eu\|_1 \ge c_{\text{fee}} r \|Ed(\theta)\|_1 + c_{\text{fee}} \sum_{j \le L-3} c_j\, s_\theta^\top E e_j$, and Cauchy–Schwarz

with weights $1 - \lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\,W_\perp(\theta)}$. Using $r \geq \sqrt{L_0(1-\gamma D)_+} \geq \sqrt{L_0}\,(1-\gamma D)_+$ and the fact that the fee is nonnegative, with $x = \sqrt{D}$,

$$L_i(v) + c_{\text{fee}}\|v\|_1 \;\geq\; L_0(\theta) + x^2 + \Big[F_{\text{dom}}(\theta)(1-\gamma(\theta)x^2)_+ - \sqrt{W_\perp(\theta)}\,x\Big]_+ \;\geq\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown $\theta$ gives (30), and $\Psi \geq 0$, $L_0(\theta) \geq c_{\text{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \leq 1$ with $\gamma F_{\text{dom}} < 1$ the bracket is a convex quadratic in $x$ with minimum $F_{\text{dom}} - W_\perp/(4(1-\gamma F_{\text{dom}}))$, which gives the closed form.

**Corollary D.5.** Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma^2_{1,t} - \sigma^2_{0,t} = \sigma^2\xi[(h_1 + h_0) * v]_t[\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x - 1 - \log x)$ with $x = \sigma^2_{0,t}/\sigma^2_{1,t}$, is quadratic in that difference. Convex impact only increases dissipation.

**Proposition 5.1.** Standard Cramér–Rao for a Gaussian mean model with Jacobian $J$. For $K = 2$ the columns of $J$ for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of $J$ is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$.

**Theorem 5.2.** Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (17) gives zero for $m \leq 2$ and $4\tau p_- p_+(\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with $N$ independent Gaussian experiments gives (19). For the permutation-invariant loss, the identity matching of $\theta_0$ and $\theta_1$ costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least $2\tau$, any estimate within $\tau$ of one economy in $d_\infty$ is farther than $\tau$ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the projection estimator of $M_3$ is unbiased with variance $\sigma_h^2/(N\|\nabla M_3 h\|^2)$, and sub-Gaussian concentration gives the sample size; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

**Theorem 5.4.** Let $(s^A, \omega^A)$ and $(s^B, \omega^B)$ be the $K$-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K - 1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the $W_1$ version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass $W$, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

**Lemma 5.6.** Let $L = 2n - 2$ and let $p^*$ be the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree $L$, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j \neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial $p$ of degree at most $L$; multiplying by $-1$ if necessary, $w_i$ has the sign of $|t_i| - p^*(t_i)$ for every $i$. Exactly $n$ weights are positive and $n$ negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x + 1)/2$ gives $n$-atom probability measures $\pi_A$ (positive part) and $\pi_B$ (negative part) on $[0, \rho]$ with equal moments of orders $0, \ldots, L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f \, d(\pi_A - \pi_B) = \frac{\rho}{2} \int \big(|x| - p^*(x)\big) \, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because $p^*$ integrates to the same value under both measures, $|x| - p^* = E$ on the support of $\pi_A$ and $-E$ on that of $\pi_B$. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \geq \rho E \geq c_* \rho / L$.

**Theorem 5.7.** Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the $\ell_1$ norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with $\pi_A, \pi_B$ from Lemma 5.6; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1-rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1} (1-rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \leq C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the $N$-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \leq (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \geq \Lambda_N - 2\log(1/\rho)$. Take $n$ the smallest integer with this property and $n \geq 2$; then $2n - 2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N \geq 4\log(1/\rho)$. Since $d$ is a norm on signed measures and $d(\mu_A, \mu_B) = W\, W_1(\pi_A, \pi_B) \geq W c_* \rho/(2n-2) > 2c_* \rho W \log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j \in \{A,B\}} P_j(d(\hat{\mu}, \mu_j) \geq c_* \rho W \log(1/\rho)/\Lambda_N) \geq \frac{1}{4} e^{-1}$.

(ii) *Approximation.* Let $f$ be 1-Lipschitz on $[0, \rho]$ with $f(0) = 0$, so $|f| \leq \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1, 1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial $p$ of degree $L$ with $\sup|g - p| \leq (\pi/2)(\rho/2)/(L+1) \leq \rho$, hence $\sup|p| \leq 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \leq 2\sup|p| \leq 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates $f$ with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \leq \sum_{j \leq L} |b_j|\, \|T_j(2r/\rho - 1)\|_{\text{coef}} \leq 4\rho \sum_{j \leq L} B(\rho)^j \leq \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\text{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\text{coef}} \leq (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of $y^i$ by at most $(1+\alpha)^i$, and $B(\rho) - 1 \geq 4$. *Moment comparison.* If $\max_{j \leq L} |\int r^j d\hat{\mu} - k_j| \leq 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\, d(\hat{\mu} - \mu) = \int (f - q)\, d(\hat{\mu} - \mu) + \sum_{j \leq L} q_j \Big( \int r^j d\hat{\mu} - k_j \Big) \leq \frac{\pi\rho(W + \delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

*Kernel recovery.* From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \dots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \dots, h_{L+1}$ only. Work in the algebra of power series modulo $z^{L+2}$ with the $\ell_1$ norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$, then $1/\hat{H} = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \le 2\|1/H\|_1^2 \|E\|_1$, so $\max_j |\hat{k}_j - k_j| \le \delta \equiv 2((1+s)/\kappa)^2 (L+2) \max_{t \le L+1} |E_t|$. Each $E_t$ is Gaussian with variance $\sigma^2/N$, so with probability at least $1 - \beta$, $\max_t |E_t| \le \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true $\mu$ is feasible for the linear program with value at most $\delta$, so the minimizer satisfies the moment condition above. *Choice of $L$.* With $L + 1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4} L \sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \le 2\pi\rho W \log B(\rho)/(\log(N/\sigma^2) - 4 \log B(\rho))$. For $N/\sigma^2$ larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W \log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

**Theorem 5.9.** For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi} \int_{-\pi}^{\pi} \big(\varrho - 1 - \log \varrho\big)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = H_1 \Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(H_1 \Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with $w_k$ the cohort weights (any factor of $\eta_k$ is absorbed in $\phi$), where $\phi$ is analytic in $\eta$ on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding $\phi$ around $\bar{\eta}$, $\Delta G = \sum_{j \ge 0} \frac{\partial_\eta^j \phi(z, \bar{\eta})}{j!} \Delta \tilde{M}_j$, where $\tilde{M}_j = \sum_k w_k (\eta_k - \bar{\eta})^j$. For the pair, $\Delta\tilde{M}_0 = \Delta\tilde{M}_1 = 0$ and $\Delta\tilde{M}_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde{M}_j = 0$ for $j \le 2$ and $\Delta\tilde{M}_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (18); for the $K$-cohort Gauss and Gauss–Radau economies with rates $\bar{\eta} + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K - 2$ exactly, so $\Delta\tilde{M}_j = W(\Delta\eta)^j \sum_k (\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for

$j \le 2K-2$ and equals $\pm W g_K(\Delta\eta)^{2K-1} \neq 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde{M}_j| \le 2W(\Delta\eta)^j$, and analyticity of $\phi$ in $\eta$ on a disk of radius $\varrho > 0$ around $\bar{\eta}$, uniformly on the circle, gives $|\partial_\eta^j \phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar{\eta})\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function $c$ does not vanish identically: $H_0\partial_\eta^j\phi$ is analytic outside the disk and non-constant, and an analytic function whose real part vanishes on the circle is an imaginary constant. For $|u| \le \frac{1}{2}$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements follow from $D(P_1^T \| P_0^T) = TD + O(1)$ for stationary Gaussian processes with smooth spectra and the Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6); attainment follows from Stein's lemma for the likelihood-ratio test.

**Theorem 5.8.** In the delay variable $\lambda = z^{-1}$, $H = \kappa \prod_i (1 - r_i\lambda) / \big[(1-a\lambda)\prod_i(1-r_i\lambda) - \kappa\lambda\sum_i w_i \prod_{j\neq i}(1-r_j\lambda)\big]$, which is (4) divided through by $z^{K+1}$; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem 5.8. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z\prod_i(z-r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is $H$; Theorem 3.1 then gives the merged measure.

**Proposition 6.1.** With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort $i$, so $\tilde{u}_{i,t} = r_i\tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i \sum_{s\ge 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives $\Sigma_i$. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.3), so $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell \sum_m \omega_m/(1-\lambda_\ell\lambda_m) \ge 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i \sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each

term positive. The scores $Z_t e_{i,t}$ are one-dependent with $\mathbb{E}[Z_t Z_t^\top e_{i,t}^2] = \sigma_f^2(1 + r_i^2)\Gamma_0$ and $\mathbb{E}[Z_t Z_{t-1}^\top e_{i,t} e_{i,t-1}] = -r_i \sigma_f^2 \Gamma_1$, so their long-run variance is $\Omega_i$, and the central limit theorem for $m$-dependent sequences gives (20). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \le \bar{\lambda} < 1$, so $\det \Sigma_i \ge w_{\min}(1 - \bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

**Theorem G.1.** With $J_\theta$ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and $P_{11}$ is affine in $s_2$ with positive slope, so for each $\theta$ a unique $s_2(\theta) \ge 0$ gives $P_{11} = V$ when $V$ exceeds the value at $s_2 = 0$. The one-step price variance is $s_1$ throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \ge |\theta|$.

# B Notation

| Symbol | Meaning |
|---|---|
| $x_t$ | pricing error (price minus fundamental) |
| $a$ | price persistence (anchor), $0 \le a < 1$ |
| $\kappa$ | price impact per unit of executed quantity |
| $q_i,\ \eta_i,\ r_i$ | cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$ |
| $w_i = q_i \eta_i$ | merged weight |
| $M = \sum_i w_i \delta_{r_i}$ | merged rate–weight measure |
| $k_m,\ G(z)$ | feedback kernel and its transform |
| $H(z),\ h_j$ | closed-loop transfer function and impulse response, $h_0 = \kappa$ |
| $J,\ S$ | state transition matrix and $S = a + \kappa \sum q_i$ |
| $v_t,\ V$ | executed probe quantity and total quantity $\sum \lvert v_t \rvert$ |
| $d_T$ | impulse-response difference between hypotheses, first $T$ taps |
| $\sigma$ | observation-noise standard deviation |
| $\delta,\ \Delta\eta$ | half-separation of a known pair; spacing of unknown cohorts |

| Symbol | Meaning |
|---|---|
| $K_*$ | transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$ |
| $T_h$ | Toeplitz matrix $(h_{\lvert i-j\rvert})$ of execution cost |
| $g_i(\omega)$ | dissipation kernel $\operatorname{Re}\hat{H}_i(e^{i\omega}) - \kappa/2$ |
| $B,\ \mu,\ c,\ \phi$ | capital, maintenance-margin fraction, proportional cost, taker fee |
| $V_{\text{eff}}$ | per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$ |
| $\hat{Q},\ \hat{S}$ | reduced-form loop gain and persistence indices |

# C Internet appendix

## C.1 A finite-window price-surveillance bound

**Theorem C.1** (Finite-window abstention)**.** *Observe $x_t = sx_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \le T$, with $s \in (0, 2)$. Let a possibly randomized rule $\Psi$ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ and $\sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$ with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. On $B_T = (1 - c_0/T, 1 + c_0/T)$,*

$$\inf_{s\in B_T} P_s(\Psi = \text{Abstain}) \ \ge\ 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

*and for every finite $T$ the minimax worst-case error of binary rules over $s \in (0, 2)$ is at least $1/2$.*

*Proof.* Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval $B_T$ is an abstention layer that any error-controlled monitor must respect in

the scalar Gaussian case; richer disclosure can remove it.

## C.2 Passive distinction of a known pair in finite samples

**Proposition C.2** (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma^2)$, use independent three-step reset blocks and no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q\eta$, $b_{2,0} = b_1^2 + \kappa q\eta(1-\eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q\delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With*

$$R = \max_{r \in \{0,\delta\}} \Big\{ (s_w^2 + \sigma^2)\big[s_w^2(1 + b_1^2 + b_{2,r}^2) + \sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

*the midpoint test of the average product has worst-case error at most $\alpha$ whenever $n \geq 4R/(\alpha\Delta^2)$ blocks are observed: zero controlled quantity and order $\delta^{-4}$ blocks.*

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for $3.9 \times 10^7$ blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with $2 \times 10^5$ blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

*Proof.* In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \leq 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \leq 0$ for uniform innovations, so it is at most $R$. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

## C.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - a h_t$.

2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.

3. *Transmission zeros:* compute the zeros of the realization, which under known $(\kappa, a)$ isolate the roots of the denominator of $G$.

4. *Weights:* with recovered rates, project $u_t = (x_t - ax_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its nonstructural transmission zeros are the memory rates (Theorem 3.1); the known timing zero is discarded, and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 5.4 and 5.5, which is why we do not report it as an estimator of individual cohorts.

### C.4 Probing protocols

### C.5 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 [RECOMPUTED IN CLOSED FORM]. The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because $z$ contains no inventory. Margin must be computed on a cash-and-inventory ledger.

## D Internet appendix: Account-level capital

Lemma 4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin

Table 16: Probing protocols at observation horizon $T = 4$ per cycle

| Protocol | $\delta$ | Executed vol. | Max trade | Error (SE) | Capital |
|---|---|---|---|---|---|
| A: single pulse (unconstrained) | 0.03 | 131.5 | 131.5 | 0.047 (0.003) | 6,985.91 |
| | 0.05 | 47.3 | 47.3 | 0.054 (0.004) | 921.73 |
| | 0.08 | 18.5 | 18.5 | 0.053 (0.004) | 146.73 |
| | 0.12 | 8.2 | 8.2 | 0.044 (0.003) | 31.45 |
| B: repeated capped pulses | 0.03 | 2,768.8 | 6.2 | 0.050 (0.003) | 7,035.75 |
| | 0.05 | 362.5 | 6.2 | 0.054 (0.004) | 923.88 |
| | 0.08 | 56.2 | 6.2 | 0.047 (0.003) | 146.00 |
| | 0.12 | 12.5 | 6.2 | 0.038 (0.003) | 34.88 |
| C: sequential test, capped | 0.03 | 1,410.2 | 6.2 | 0.045 (0.003) | 3,585.14 |
| | 0.05 | 195.6 | 6.2 | 0.038 (0.003) | 500.00 |
| | 0.08 | 35.9 | 6.2 | 0.032 (0.003) | 94.30 |
| | 0.12 | 10.5 | 6.2 | 0.011 (0.002) | 29.85 |
| D: generalized eigenvector | 0.03 | 2,359.7 | 3.7 | 0.050 (0.003) | 2,632.77 |
| | 0.05 | 314.8 | 3.7 | 0.048 (0.003) | 353.67 |
| | 0.08 | 49.2 | 3.7 | 0.046 (0.003) | 57.53 |
| | 0.12 | 9.9 | 3.8 | 0.045 (0.003) | 13.56 |

2,000 replications per hypothesis and cell. Log–log capital slopes: −3.90, −3.84, −3.47, −3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D. Kind: [RECOMPUTED] (`recompute_p1_protocols.py`).

dollars.

**Numerical status.** The analytic inequalities below are conditional mathematical bounds. The displayed decimal optima and brackets are numerical evaluations, not independently outward-certified enclosures of the continuous frequency or angle extrema. A finite grid and a matching archived number do not establish that enclosure; rigorous optimization remains a separate verification obligation.

## D.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates $\theta_0$ and $\theta_1 = \theta_0 + \delta$ over $N$ periods, submitting probes $V = (V_1, \ldots, V_N)^\top$ that generate inventory $I_t = \sum_{s \le t} V_s$. Under contract $I_0$ (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{22}$$

The observer minimizes capital $C(N,V) = \sum_t(\lambda V_t^2 + \phi|V_t|) + \max_t \mu p_0|I_t|$ subject to (i) separation $D_{\mathrm{KL}} \geq \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \leq V_{\mathrm{eff}}$ and $|I_t| \leq I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \leq B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

**Proposition D.1** (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon $T$ per pulse, $(\delta, B)$ space splits into three regions.*

*(i)* Feasible single pulse. *For $\delta \geq \delta_{\min}(T)$ and $B \geq C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

*(ii)* Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds $V_{\mathrm{eff}}$. Capped pulses each supply divergence $V_{\mathrm{eff}}^2\|d_T\|^2/(2\sigma^2)$, so*

$$N_{\mathrm{pulses}}(T,\delta) = \left\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\mathrm{eff}}^2 \|d_T\|_2^2} \right\rceil. \tag{23}$$

*(iii)* Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\mathrm{session}})$ is the root of $V_{\mathrm{eff}}\|d_{T_{\mathrm{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha/\sqrt{\lfloor T_{\mathrm{session}}/T_{\mathrm{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\mathrm{session}}^{-1/4})$.*

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\mathrm{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\mathrm{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\mathrm{eff}}^2 + \phi P_* V_{\mathrm{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving

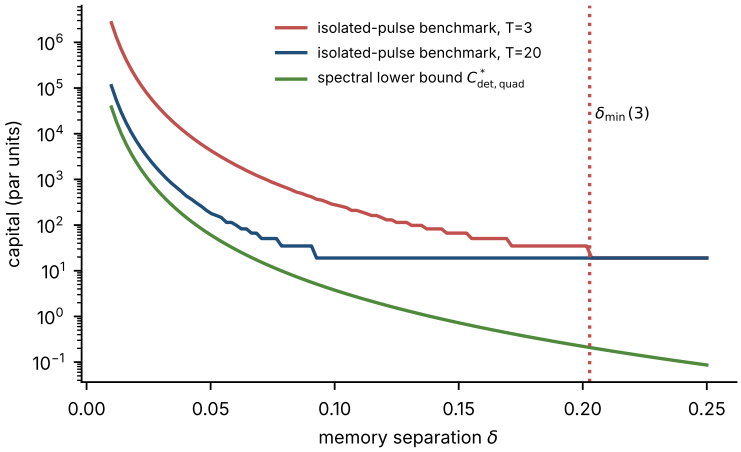



Figure 4: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem D.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

Table 17: Account-level initial capital at $\delta = 0.05$ (par units)

| Design | $C_0$ | Error | End inv. | Peak margin | Max $\lvert v_t \rvert$ | Trades |
|---|---|---|---|---|---|---|
| 12 buys, $T = 20$ apart, never unwound | 229.13 | 0.0578 | 75.00 | 39.38 | 6.25 | 12 |
| 6 buy/sell round trips | 189.45 | 0.0308 | 0.00 | 3.28 | 6.25 | 12 |
| 5 buy/sell round trips | 157.89 | 0.0442 | 0.00 | 3.28 | 6.25 | 10 |
| 12 buys, each unwound after 20 steps | 378.82 | 0.0040 | 0.00 | 3.28 | 6.25 | 24 |
| Generalized-eigenvector probe ($L = 120$) | 65.06 | 0.0500 | 0.00 | 2.74 | 1.24 | 120 |
| Generalized-eigenvector probe ($L = 160$) | 64.86 | 0.0500 | 0.00 | 2.17 | 1.09 | 160 |
| Fee-refined probe ($L = 240$) | 64.48 | 0.0500 | 0.00 | — | 1.30 | 240 |

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The numerically evaluated lower bound of Theorem D.2 is 61.41; the explicit constant of Lemma 4.5 is $4.9 \times 10^{-4}$.

comes from observing the transient response rather than a single tap (Figure 4).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

## D.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table 17 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.25 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{24}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon $L$ into $L+1$ at identical cost and information, so the minimum is non-increasing in $L$.

**Theorem D.2** (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{25}$$

*(i)* Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

*(ii)* Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most $\alpha$ has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\text{det,quad}}(\delta). \tag{26}$$

*(iii)* Bracket. *At $\delta = 0.05$, $C^*_{\text{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the numerical lower estimate $C_{\text{det}} \geq 61.41$. A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

*(iv)* Process noise. *Let the price also receive the innovations $w_t \sim N(0, \sigma_w^2)$ of (1), inde-*

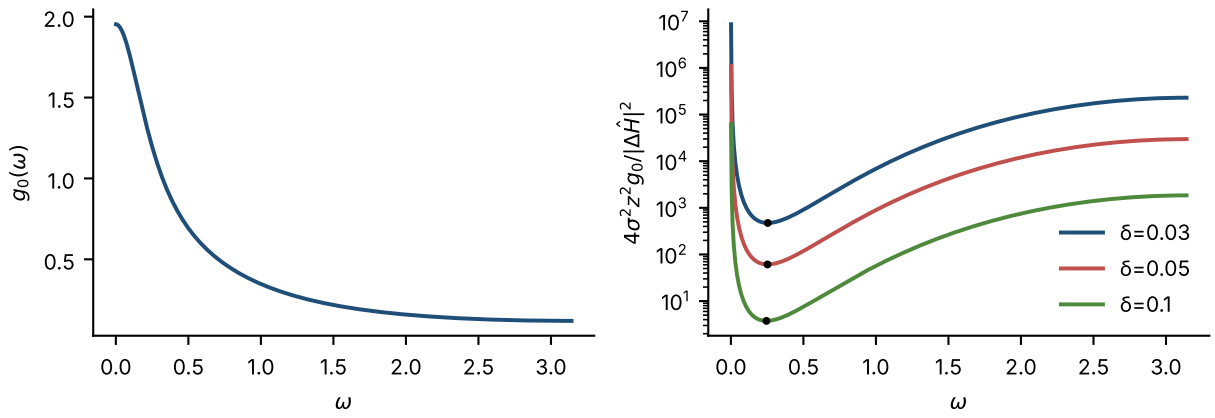



Figure 5: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (26); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

*pendent of observation noise* $\nu_t \sim N(0, \sigma_\nu^2)$*, so that from the reset state the observer sees* $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$*,* $t < T$*. For the known pair of Theorem 4.1, put* $\sigma_{\text{eff}}^2 = \sigma_\nu^2 + \sigma_w^2/4$*,* $\bar{M} = 1/(1-S)$ *and* $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2(\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma_{\text{eff}}^4)$*. Every deterministic round trip whose test has errors* $\alpha$ *and* $\beta$ *under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma_{\text{eff}}^2 \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{27}$$

*with* kl *as in Theorem D.3. In-loop noise therefore leaves the* $\delta^{-4}$ *capital order intact for every horizon* $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$*; at horizons of order* $\delta^{-4}$ *and longer, the noise itself separates the pair without a probe (Theorem 5.9).*

Positivity of $g_i$ follows from Lemma 2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\text{det,quad}}$ across separations (Table 18): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\text{det,quad}}$ between $3.67\times10^{-4}$ and $3.84 \times 10^{-4}$, confirming the $\delta^{-4}$ law. The fee term, the feasible upper bounds and the adaptive bound of Theorem D.3 are reported in the same table.

**Theorem D.3** (Adaptive and sequential probing)**.** *Let* $(v_t)_{t\leq\tau}$ *be any non-anticipating policy adapted to the price filtration, with an almost surely finite stopping time and* $\sum_{t\leq\tau} v_t = 0$*.*

Table 18: Capital lower bounds and feasible upper bounds across separations (par units)

| $\delta$ | $\omega^*$ | $C^*_{\mathrm{det,quad}}$ | $\delta^4 C^*_{\mathrm{det,quad}}$ | Certified $C_{\mathrm{det}}$ | Feasible UB | Adaptive LB |
|---|---|---|---|---|---|---|
| 0.02 | 0.2546 | 2397.85 | $3.84 \times 10^{-4}$ | 2399.58 | 2428.25 | 1176.04 |
| 0.03 | 0.2541 | 473.19 | $3.83 \times 10^{-4}$ | 473.97 | 484.50 | 232.51 |
| 0.05 | 0.2526 | 61.14 | $3.82 \times 10^{-4}$ | 61.41 | 64.48 | 30.22 |
| 0.08 | 0.2488 | 9.26 | $3.79 \times 10^{-4}$ | 9.37 | 10.38 | 4.64 |
| 0.10 | 0.2453 | 3.77 | $3.77 \times 10^{-4}$ | 3.83 | 4.44 | 1.91 |
| 0.15 | 0.2331 | 0.73 | $3.67 \times 10^{-4}$ | 0.76 | 1.01 | 0.38 |

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. From `recompute_theory.py` and `recompute_p1_capital.py`; the numerical-bound column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \leq 240$; the adaptive column adds the fee lower bound to the adaptive bound.

*Then*

$$\mathbb{E}_0[L_0(v)] \;\geq\; 2\sigma^2 \,\mathrm{kl}(\alpha, 1-\beta) \inf_{\omega} \frac{g_0(\omega)}{|\Delta \hat{H}(e^{i\omega})|^2}, \tag{28}$$

*where* $\mathrm{kl}(\alpha, 1-\beta) = \alpha \log \frac{\alpha}{1-\beta} + (1-\alpha) \log \frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free loss by at most the factor* $\mathrm{kl}(\alpha, 1-\beta)/(2z^2_{1-\alpha}) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, and cannot alter the* $\delta^{-4}$ *order.*

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a numerical ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under $H_0$, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

**Theorem D.4** (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of* $L \geq 4$ *trading periods, let* $E \in \mathbb{R}^{L \times (L-1)}$ *be the first-difference basis of flat probes (*$\sum_t v_t = 0$*),* $Q_i = \frac{1}{2} T_{h_i}$ *(so that* $L_i(v) = v^\top Q_i v$*),* $A_{\mathrm{diff}} = (H_1 - H_0)^\top (H_1 - H_0)$*, and let* $G_i = E^\top Q_i E \succ 0$ *and* $M = E^\top A_{\mathrm{diff}} E \succeq 0$ *be the reduced execution-loss and information matrices, with* $G_i$*-orthonormal generalized eigenvectors* $e_j$ *and eigenvalues* $0 \leq \lambda_1 \leq \cdots \leq \lambda_{L-3} < \lambda_{L-2} \leq \lambda_{L-1}$*; put*

$c_{\mathrm{req}} = 4\sigma^2 z_{1-\alpha}^2$ *and* $c_{\mathrm{fee}} = \phi P_*$. *For* $\theta \in [0,\pi)$ *let* $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$, $\lambda(\theta) = \lambda_{L-2}\cos^2\theta + \lambda_{L-1}\sin^2\theta$, $s_\theta = \mathrm{sign}(Ed(\theta))$, *and*

$$F_{\mathrm{dom}}(\theta) = c_{\mathrm{fee}}\sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\,\|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c_{\mathrm{fee}}^2 \sum_{j\le L-3} \frac{(s_\theta^\top E e_j)^2}{1-\lambda_j/\lambda(\theta)},$$
$$\gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta)-\lambda_{L-3})\,c_{\mathrm{req}}}. \tag{29}$$

*Then every flat probe meeting the separation constraint of* (24) *satisfies*

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; \min_\theta\Big\{\frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta)\Big\} \;\equiv\; C_{\mathrm{RH}}(L) \;\ge\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{30}$$

$$\Psi(\theta) = \min_{x\ge 0}\Big\{x^2 + \Big[F_{\mathrm{dom}}(\theta)\big(1-\gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\,x\Big]_+\Big\} \;\ge 0,$$

*a piecewise-quadratic minimization in closed form; when* $\gamma F_{\mathrm{dom}} < 1$ *and the minimizer has* $\gamma x^2 \le 1$, $\Psi = \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1-\gamma F_{\mathrm{dom}}))\big)$.

The bound is computable for any horizon. Table 19 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective, so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or $L_1$-localized probes are needed at long horizons.

## D.3 Robustness: heteroskedastic impact and probing protocols

**Corollary D.5** (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma_{i,t}^2(v) = \sigma^2(1+\xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\mathrm{eff}} = \kappa(1+\xi_0|v_t|)$. *For capacity-bounded probes,* $D_{\mathrm{KL}}(P_0^v\|P_1^v) \le \frac{1+O(\xi^2 V_{\mathrm{eff}}^2)}{2\sigma^2}\|(H_1-H_0)v\|_2^2 = O(\|v\|_2^2\delta^4)$, *and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$.

Table 19: Finite-horizon capital at $\delta = 0.05$ (par units)

| Horizon $L$ | Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$ | Full bound $C_{\mathrm{RH}}(L)$ | Best feasible probe | Gap to $C_{\mathrm{det}} = 61.41$ |
|---|---|---|---|---|
| 80 | 62.55 | 64.72 | 65.02 | 5.9% |
| 120 | 61.84 | 63.39 | 64.59 | 5.2% |
| 160 | 61.54 | 62.46 | 64.50 | 5.0% |
| 240 | 61.32 | 61.76 | 64.48 | 5.0% |
| 320 | 61.25 | — | $\leq 64.48$ | $\leq 5.0\%$ |
| 480 | 61.19 | — | $\leq 64.48$ | $\leq 5.0\%$ |

Quadratic bound $c_{\mathrm{req}}/\lambda_L$ from the generalized eigenproblem of Theorem D.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with $\lambda'$ computed against $T_{h_0}$ rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (30) over 3,600 values of $\theta$ (`hardening/rh_bound.py`); at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near $-4$ in $\log \delta$: $-3.90$, $-3.84$, $-3.47$ and $-3.81$ over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table 16). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

## D.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2\delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\mathrm{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about $9.9 \times 10^4$ USDT, which is not prohibitive for a well-capitalized desk (Table 20). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 5.4, when several

Table 20: Capital lower bound under alternative observation-noise calibrations (USDT)

| Noise calibration | $\delta = 0.02$ | $\delta = 0.05$ | $\delta = 0.10$ |
|---|---|---|---|
| Roll estimator, median (0.40 USDT) | 3,874,696 | 99,193 | 6,200 |
| One-second price-change s.d., median (1.98 USDT) | 93,688,369 | 2,398,422 | 149,902 |
| Earlier assumption (4 USDT, upper scenario) | 383,812,500 | 9,825,600 | 614,100 |

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with $\sigma$ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Table 21: Daily observation-noise estimates, Binance BTCUSDT one-second closes

| Date | Roll $\hat{\sigma}$ (USDT) | One-second change s.d. (USDT) | Mean price (USDT) |
|---|---|---|---|
| 2023-01-10 | 0.152 | 0.679 | 17,288 |
| 2023-01-11 | 0.092 | 0.891 | 17,456 |
| 2023-01-12 | 0.232 | 2.292 | 18,343 |
| 2023-01-13 | 0.348 | 1.762 | 19,075 |
| 2023-01-14 | 0.773 | 3.511 | 20,818 |
| 2023-01-15 | 0.482 | 1.790 | 20,778 |
| 2023-01-16 | 0.629 | 2.357 | 21,036 |
| 2023-01-17 | 0.457 | 1.976 | 21,194 |
| 2023-01-18 | 0.695 | 2.541 | 21,122 |
| 2023-01-19 | 0.402 | 1.316 | 20,850 |
| 2023-01-20 | 0.316 | 2.038 | 21,306 |
| Median | 0.402 | 1.976 | |

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

cohorts are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

# E Internet appendix: Near-colliding rates: a uniform rate for the merged measure

Theorem 5.5 needs the rates to stay $\gamma$-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights $w$ and $W - w$ at rates $r$ and $r + \Delta$ and two of weights $W - w$ and $w$ at the same rates have merged measures

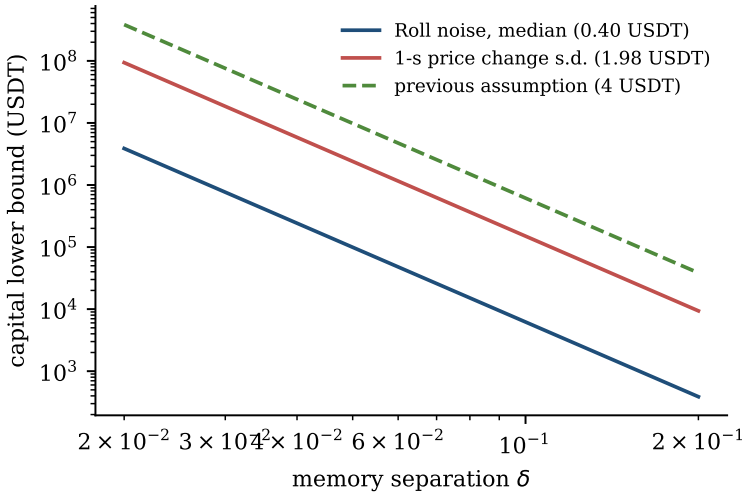



Figure 6: Capital lower bound against memory separation for three observation-noise calibrations.

that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

**Theorem E.1** (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most $K$ atoms in $[0, \bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From $N$ reset experiments as in Theorem 5.5, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is $C$ depending only on $(K, \bar{r}, a, \kappa, B)$ such that, with probability at least $1 - \beta$,*

$$W_1(\hat{\bar{M}}, \bar{M}) \;\leq\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \text{uniformly over } \mathcal{M}_K.$$

*Hence $N = O(\sigma^2 m_{\min}^{-2} \epsilon^{-(4K-2)})$ experiments give accuracy $\epsilon$, and with the Wasserstein statement of Theorem 5.4 the minimax rate for the merged memory distribution is $N^{-1/(4K-2)}$, uniformly over near-colliding economies.*

*Proof.* As in the proof of Theorem 5.5, the kernel values $k_0, \dots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \le e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$. The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \le 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \le k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy $|m_t(\hat{M}) - m_t(M)| \le 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \le 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries both normalized measures to distributions on $[-1,1]$ with at most $K$ atoms, multiplies $W_1$ by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1+2/\bar{r})^{2K-1}$. The moment comparison inequality for $K$-atomic distributions on $[-1,1]$ (Wu and Yang, 2020), $W_1 \le c_K\delta^{1/(2K-1)}$ when the first $2K-1$ moments differ by at most $\delta$, then gives the bound with $C = c_K(\bar{r}/2)(1+2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem 5.5 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes impossible as rates collide, at any sample size. A supervisor who needs the first can have it at the rate of Theorem 5.4; one who needs the second has to be told the number of cohorts and a separation, or has to obtain tagged data (Section 6).

# F Internet appendix: Robustness to nonlinear and adaptive trading rules

The theorems of Sections 3–5 use the linear exponential-memory recursion (2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size $A$ moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and

locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1 - \eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size $\eta_i$; and (5) a *tabular value rule* that updates a 21-bin table at rate $\eta_i$, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate $\eta_i$, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight $q$ and position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B ($\delta^{-4}$ law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as $\delta^2$ and the certified capital of Theorem D.2 as $\delta^{-4}$. The saturation threshold $A_{\text{sat}}$ is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-registered a second experiment (`P1_trained_learners.json`, SHA-256 `3e8bf30cbfb6...`) in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with

forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate $\eta_i$: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and $\delta^{-4}$ predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative $L_2$ response at $A = 0.005$; (c) the certified capital to separate trained populations whose memory rates differ by $2\delta$ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the $\delta^{-4}$ law fail when learning continues during the probe. A first implementation of this experiment did not pass the trained parameters to the probing simulator; the results below come from the corrected implementation (`validate_trained_learners_hardened.py`), which reloads the saved parameters.

Table 22 and Figure 7 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and certified capital scale as $\delta^2$ (slope 2.007) and $\delta^{-4}$ (slope $-4.001$), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most $10^{-5}$ in relative response at $A = 0.005$, and certified capital scales with slopes $-4.014$ to $-3.973$, inside the registered band. The level of capital differs across learners because training sets the response gain.

Table 22: Merged-measure equivalence and the $\delta^{-4}$ law under stylized nonlinear rules and closed-loop trained online learners

| Rule | Test A raw ($O(A^2)$) | Test A norm. ($O(A)$) | Test B slope ($\|\Delta h\|_2$) | Test B slope (capital) | Capital at $\delta = 0.05$ | $A_{\text{sat}}$ / $R^2_{\text{exp}}$ |
|---|---|---|---|---|---|---|
| *Panel A. Stylized nonlinear and adaptive rules (fixed operating point)* | | | | | | |
| Linear benchmark | — | — | 2.007 | −4.001 | 61.42 | — |
| 1. Momentum (tanh limit) | 2.015 | 1.015 | 2.007 | −4.001 | 61.47 | 0.834 |
| 2. Threshold rule, averaged | 2.043 | 1.043 | 2.007 | −4.001 | 61.42 | 0.049 |
| 3. Adaptive gain (RLS) | 2.064 | 1.064 | 2.007 | −4.001 | 61.46 | 0.751 |
| 4. Online neural readout | 2.011 | 1.011 | 2.007 | −4.001 | 61.48 | 0.751 |
| 5. Tabular value, smoothed | 2.039 | 1.039 | 2.007 | −4.001 | 61.44 | 1.271 |
| *Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)* | | | | | | |
| 6. Recursive least squares | 3.000 | 2.000 | 2.008 | −4.014 | 54.58 | $R^2 = 0.993$ |
| 7. Two-layer network (SGD) | 1.878 | 0.878 | 2.007 | −4.012 | 65.19 | $R^2 = 0.994$ |
| 8. Policy gradient on profit | 3.000 | 2.000 | 2.004 | −4.006 | 58.05 | $R^2 = 0.995$ |
| 9. Tabular Q-learner (TD(0)) | 2.108 | 1.108 | 1.994 | −3.973 | 186,821 | $R^2 = 1.000$ |
| *Panel C. Where the trained-learner results change (range over the four learners)* | | | | | | |
| Independent training streams | rel. gap $\leq 10^{-5}$ | | 0.13 to 0.92 | −1.75 to −0.38 | — | (c) fails |
| Learning during the probe | rel. gap $\leq 1.1 \times 10^{-3}$ | | 1.99 to 2.01 | −4.02 to −3.98 | — | (c) holds |
| Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past $\tau$ only for $A > 0.139$. | | | | | | |
| Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$. | | | | | | |

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2 / A$ ($O(A)$, plotted in Figure 7). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most $10^{-5}$. A raw Test A slope above 2 means the gap is of higher order than $A^2$. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, certified bound of Theorem D.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital. Kind: recomputed (Panel A: `recompute_p1_learners.py` → `R_P1_learners.json`; Panels B–C: `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json`).

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are −1.75, −0.97, −1.62 and −0.38. Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The $\delta^{-4}$ law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during

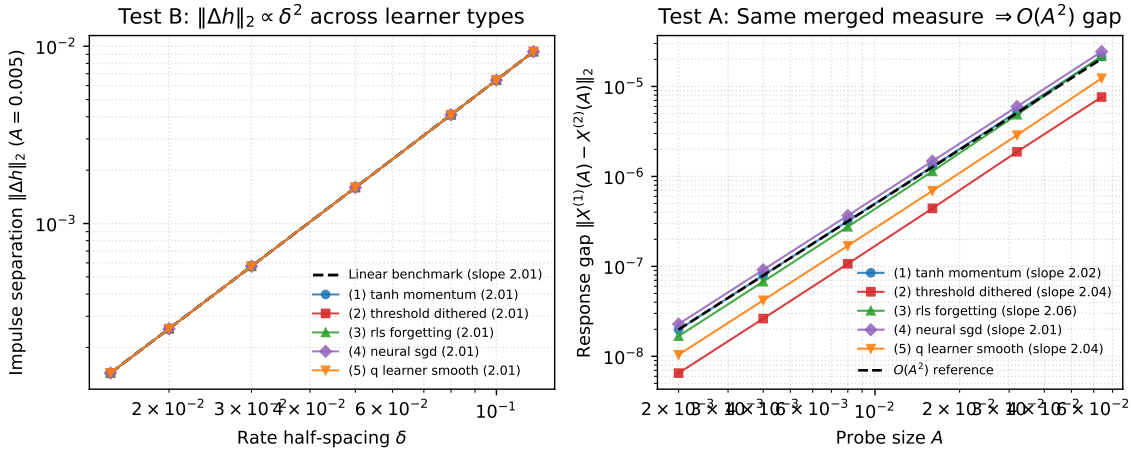



Figure 7: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing $\delta$ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size $A$, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

the probe at rate 0.35, the law survives (slopes $-4.02$ to $-3.98$) and the merged-measure gap rises only to $1.1 \times 10^{-3}$, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights, not their rates.

# G Internet appendix: A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

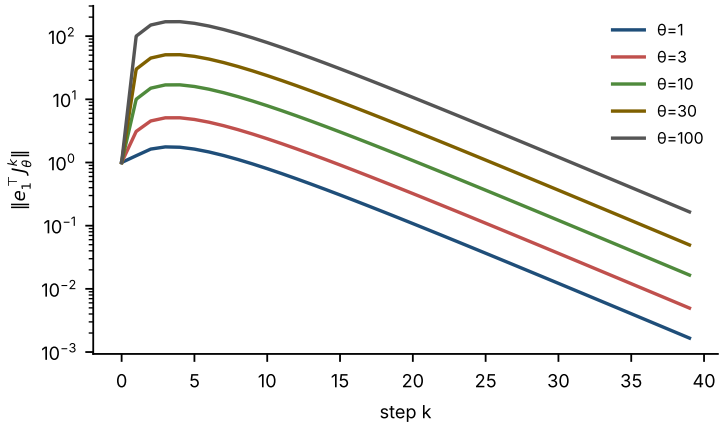



Figure 8: Price-row transient gain of the spectral-blindness family $J_\theta$: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every $\theta$.

**Theorem G.1** (Spectral blindness)**.** *For any eigenvalues* $0 < \lambda_2 < \lambda_1 < 1$*, one-step price variance* $s_1 > 0$ *and stationary price variance* $V$ *large enough, there is a family of two-state markets* $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ *with shock covariance* $\mathrm{diag}(s_1, s_2(\theta))$ *such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain* $\sup_k \|e_1^\top J_\theta^k\|$ *grows without bound in* $\theta$*.*

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure 8).

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix C.5). Second, in a network of $M$ identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator

(`recompute_oct20.py`). A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial impact, which also depends on the input and output maps. The positive-cohort no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

# H Internet appendix: Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 6.4 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

## H.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces $p_t^*$ by a trailing filter. Let $p_b$ be a log price proxy on uniform bars of length $\Delta$ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{31}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{32}$$

Table 23: Samples

| Sample | Coverage | Size | Source |
|---|---|---|---|
| Binance spot panel, 8 pairs | 96 dates, 2021–2024 | 66.3M one-second bars | Binance archive |
| Binance spot fee-event windows, 8 pairs | ±20 days around 2022-07-08 and 2023-03-22 | 640 pair-days | Binance archive |
| Binance USD-M perpetuals, 30 symbols | 90 dates, 2023–2026 | 2,565 symbol-days | Binance archive |
| Binance BTCUSDT one-second panel | 72 dates, 2021–2023 | 6.2M one-second bars | Binance archive |
| Binance BTCUSDT noise-calibration days | January 10–20, 2023 | 11 days | Binance archive |

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

the filtered state is an AR(1) with mechanical persistence $a_0 = 1 - \alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let $f_t$ be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{33}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1 - \hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1 - \hat{J}_{11}}. \tag{34}$$

Equation (32) implies that $\hat{J}_{11}$ must be compared with $1 - \alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

## H.2 Data

## H.3 Clock alignment dominates reduced-form feedback

We estimate (33) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions

Table 24: Multi-year Binance spot panel: reduced-form indices under three timing constructions

| Construction | $\hat{Q}$ | Date interval | $\hat{S}$ | Date interval | $\hat{Q} > 0$ |
|---|---|---|---|---|---|
| Forward-paired flow (look-ahead) | 0.0081424 | [0.0072965, 0.0089882] | 0.92074 | [0.91665, 0.92482] | 87.4% |
| Observed-flow EMA | 0.0003733 | [0.0002351, 0.0005116] | 0.92072 | [0.91640, 0.92504] | 62.4% |
| Observed, 30 flow lags | 0.0002580 | [0.0001415, 0.0003745] | 0.91738 | [0.91167, 0.92308] | 59.1% |
| Observed raw return | −0.0001555 | [−0.0003732, 0.0000622] | −0.04051 | [−0.05078, −0.03025] | 49.2% |
| Placebo random walk | 0.0000016 | [−0.0000020, 0.0000052] | 0.93510 | [0.93485, 0.93534] | 51.0% |

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

Table 25: Asset means under forward-paired and observed flow

| Pair | Forward-paired $\hat{Q}$ | Observed $\hat{Q}$ | $\hat{Q}_{30}$ | Observed $\hat{S}$ |
|---|---|---|---|---|
| BTC/USDT | 0.0141870 | 0.0006073 | 0.0004891 | 0.93464 |
| ETH/USDT | 0.0130885 | 0.0005675 | 0.0002549 | 0.93511 |
| SOL/USDT | 0.0087763 | 0.0002129 | 0.0001100 | 0.91966 |
| ADA/USDT | 0.0067465 | 0.0003928 | 0.0003546 | 0.91452 |
| DOGE/USDT | 0.0058956 | 0.0002464 | 0.0001432 | 0.92496 |
| BNB/USDT | 0.0057420 | 0.0004479 | 0.0002762 | 0.89533 |
| XRP/USDT | 0.0057034 | 0.0002249 | 0.0001508 | 0.92362 |
| LTC/USDT | 0.0049997 | 0.0002870 | 0.0002853 | 0.91792 |

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

(Table 24), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

## H.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state $x_t$ is the log price deviation from its EMA (span 30 one-second bins), $f_t$ is next-second signed taker flow, and a VAR(1)

Table 26: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

| | Real prices | Placebo (independent random walk) |
|---|---|---|
| Mean loop gain $\hat{Q}$ | 0.0045 | 0.0000009 |
| 95% CI (day bootstrap) | [0.0034, 0.0056] | [−0.0000006, 0.0000023] |
| Mean price persistence $\hat{J}_{11}$ | 0.9364 | 0.9357 |
| Mechanical value $1 - 2/31$ | | 0.9355 |

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table 24.

gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table 26 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table 24; the placebo shows only that the filter does not produce it.

## H.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta\Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock $z_t$. Lagged returns are correlated with $u_{t-1}$, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move $y$ through prices but are independent of $u$ ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table 27).

OLS is badly biased. Cochrane–Orcutt with the true $\rho$, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated $\rho$ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market.

Table 27: Feedback estimators under order splitting (simulation, 500 replications)

| Estimator | Coverage at $\gamma = 0.2$ | Rejection at $\gamma = 0$ | Bias | RMSE |
|---|---|---|---|---|
| OLS | 0.000 | 1.000 | 0.892 | 0.893 |
| Cochrane–Orcutt, true $\rho$ | 0.944 | 0.054 | 0.001 | 0.039 |
| Cochrane–Orcutt, estimated $\rho$ | 0.916 | 0.080 | 0.002 | 0.043 |
| 2SLS | 0.938 | 0.044 | −0.002 | 0.116 |
| LIML | 0.938 | 0.044 | −0.003 | 0.117 |

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future $z$) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from $3.15 \times 10^{-4}$ to $3.21 \times 10^{-4}$ (Part 13, `R7`, post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table 25 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table 25, column $\hat{Q}_{30}$), consistent with persistent order splitting.

## H.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

Table 28: Zero fees on BTCUSDT: activity and measured feedback

| | Zero fees introduced (2022-07-08) | Zero fees removed (2023-03-22) | Pooled |
|---|---|---|---|
| Log quote volume, $\hat{\beta}$ | 0.945 (0.074) | 1.599 (0.080) | 1.272 (0.084) |
| Observed-flow $\hat{Q}$, $\hat{\beta}$ | 0.00058 (0.00032) | −0.00091 (0.00060) | −0.00016 (0.00036) |
| $p$-value | 0.078 | 0.140 | 0.650 |
| Pair-days | 320 | 320 | 640 |

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

### H.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\text{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table 28).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is $-0.00016$ with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

## I Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part $N$" below and in table notes refers to notebook

$N$ (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

| Result | Source | Kind |
|---|---|---|
| Complete monotonicity, 892 economies | `L26_complete_monotonicity` | recomputed |
| $d_2 = -\kappa^2 q \delta^2$ | `P1.7-T1_third_tap` | recomputed |
| Ledger identity; $T_h$ positive definite | `R2-T1`, `R2-T3` | recomputed |
| Figure 2 | `make_figs2.py` | recomputed |
| $\delta_{\min}(3)$, Table 18 recomputed columns, 61.14, 29.94 | `recompute_theory.py` → `R_P1_theory_constants.json` | recomputed |
| Pulse benchmarks (271, 58, 12, 11 pulses), Figures 1–4 | `make_figs3.py` → `R_P1_pulse_benchmarks.json` | recomputed |
| Table 3, Table 6, worked example, passive simulation | `recompute_extra.py` → `R_P1_extra.json` | recomputed |
| Daily noise (Table 21) | `P1.3-T1_sigma_obs` | real data |
| Table 5 recomputed columns | same | recomputed |
| $K$-cohort slopes (Table 7) | `T1_K_cohort_rate` | recomputed |
| ESPRIT slope −6.15 | `P1.2-T1_esprit_rate` | recomputed |
| Minimum phase, 1,733 economies; rank 6 of 6 | `T2a`, `T2b` | recomputed |
| Passive slopes −4.21, −5.93 | `T3_passive_rate` | recomputed |
| Tagged-flow slope 0.011 | `T4_disclosure` | recomputed |
| Spectral blindness gains | `T5_spectral_blindness` | recomputed |

| Result | Source | | Kind |
|---|---|---|---|
| Noise and capital grid (Table 20) | `P1.3-T1_sigma_obs` | | real data |
| Loop gain and placebo (Table 26) | `P1.6-T1_ema_placebo` | | real data |
| Estimator table (Table 27) | `P1.5-T1`, `T2`, `T3` | | recomputed |
| Data integrity (111 files) | `R2-T9_data_integrity` | | audit |
| Tables 17, 18 (certified, feasible, adaptive columns) | `recompute_p1_capital.py` `R_P1_capital_ledger.json` | → | recomputed |
| Table 10; resolution floors; Table 16 | `recompute_p1_protocols.py` `R_P1_protocols.json` | → | recomputed |
| SPRT capital figures | `recompute_p1_sprt.py` `R_P1_sprt.json` | → | recomputed |
| Table 4 ($N^*$, $\sigma_{\text{crit}}$) | `recompute_crb.py` `R_P1_crb.json` | → | recomputed |
| Table 4 (ML columns), Table 5 (estimators) | `recompute_p1_estimators.py` | | recomputed |
| Symmetric multi-asset network (Appendix G) | `recompute_oct20.py` | | recomputed |
| Proof audit: $K_*$ bound and $d_2$ identity (Theorem 4.1), moment gaps (Theorem 5.2), spectral-blindness example (Theorem G.1) | `recompute_audit.py` `R_audit_oct2026.json` | → | recomputed |
| Theorem 5.5, Table 8 | `kcohort_upper.py` `R_P1_kcohort_upper.json` | → | recomputed |
| Lem. 5.6, Thm 5.7, Tab. 9 | `general_minimax.py` | | recomputed |
| Theorem 6.2 numbers | `disclosure_value.py` `R_P1_disclosure_value.json` | → | recomputed |

| Result | Source | Kind |
|---|---|---|
| Tables 24, 25, 28 | Part 8: `V2_spot_panel`, `V3_fee_events` | real data |
| Table 13, top panel | Part 4: `X9_disclosure_metrics` | real data |
| Table 13, bottom panel; class-position forecasts | Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…) | real data |
| Table 12 | Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…) | real data |
| Basis-trade signature; real-data Cochrane–Orcutt | Part 13: `R6`, `R7` | real data (post hoc) |
| Feedback across 30 perpetuals | Part 4: `X3_T6_cross_section` | real data |
| Theorem D.4 full bound (Table 19), direct minimization | `hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json` | recomputed |
| Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant | `hardening/p1_checks.py` → `R_P1_hardening_checks.json` | recomputed |

| Result | Source | Kind |
|---|---|---|
| Stylized nonlinear and trained online rules (Table 22, Figure 7) | `recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…) | recomputed |
| Disclosure in dollars (Table 11) | `recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json` | recomputed |
| Out-of-sample post-shock return test (Section 6.4) | `recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…) | real data |
| Out-of-sample cohort-state trajectory test (Table 14) | `recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…) | real data |